Model fitting to noisy image data

models

- parametric
  - **m** = [m1, m2, …, mh]

- e.g.,
  - Lines l
  - Conics C
  - Homographies H
  - Fundamental matrixes F

data

- **X** = {x1, x2, ---, xN}
- e.g., image point features, pairs of corresponding image points
- affected by noise:
  - errors in feature extraction
- containing outliers:
  - false positives,
  - noise artifacts,
  - true data but associated to other models (pseudo-outliers)

data – model relationship

A datum x_i is

- correctly associated to a model **m**

→ x_i is an inlier to model **m**

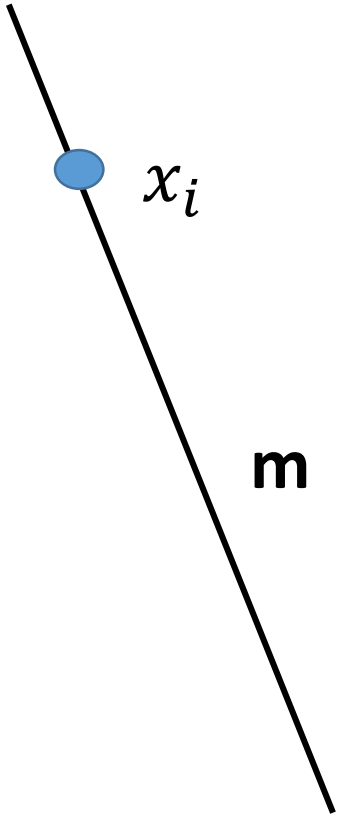
- not affected by noise

$$\boldsymbol{f}(m, x_i) = 0$$

compatibility equation

(e.g., $l^T x = 0$

$x' - Hx = 0$)



data – model relationship

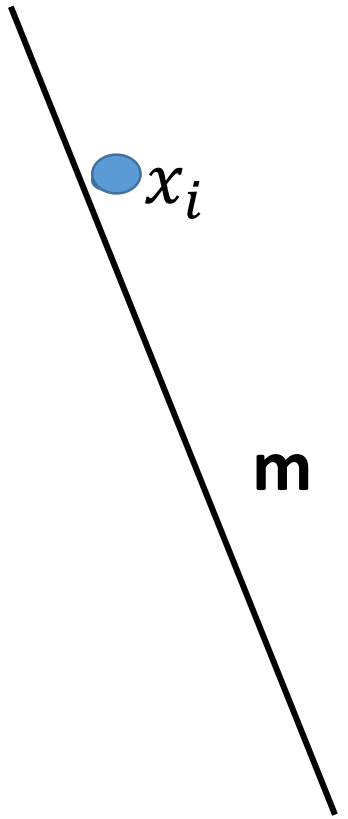
A datum x_i is

- correctly associated to a model **m**

→ x_i is an inlier to model **m**

- affected by noise

$$|\boldsymbol{f}(m, x_i)| < \epsilon \quad \text{(algebraic error)}$$



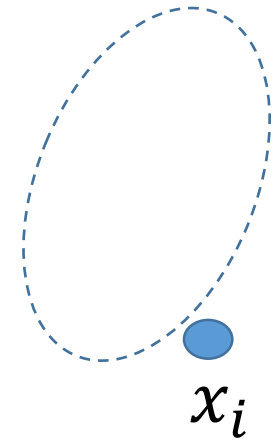
data – model relationship

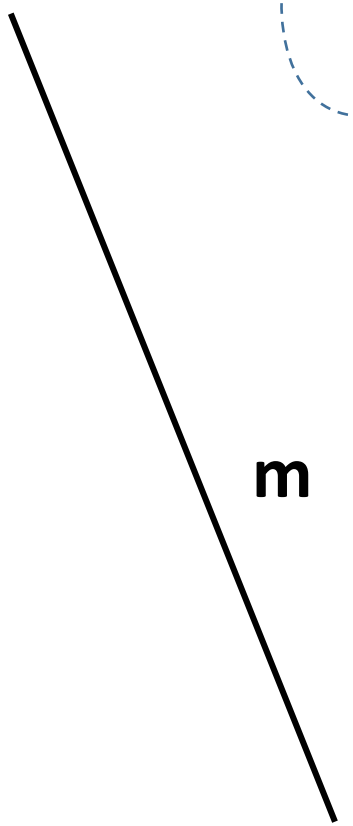
A datum x_i is

- NOT correctly associated to a model **m**

→ x_i is an **outlier** wrt model **m**

- affected by noise





$|\boldsymbol{f}(m, x_i)| \gg 0$

the *residual*

robustness

good performance in the presence of OUTLIERS

difficulty

No a priori knowledge about
which data are INLIERS and
which data are OUTLIERS

Model fitting to noisy images, simple case: NO OUTLIERS

L^k technique

$$arg\min_{m} J = \frac{1}{N}\sum_{1}^{N} \|f(m, x_i)\|^k$$

$$k = 2$$

$$\min_{m} J = \frac{1}{N} \sum_{1}^{N} \| f(m, x_i) \|^2$$

→least squares

$$k = 1$$

$$\min_{m} J = \frac{1}{N} \sum_{1}^{N} \| f(m, x_i) \|$$

→ least sum of modules

$$k = \infty$$

$$\min_{m} J = \frac{1}{N} \sum_{1}^{N} \| f(m, x_i) \|^{\infty}$$

→ min max

$$\min_m J = \frac{1}{N}\sum_1^N \|f(m, x_i)\|^{\infty}$$

multiply and divide by $(\max_i \|f(m, x_i)\|)^{\infty}$

→

$$\min_m J = (\max_i \|f(m, x_i)\|)^{\infty} \frac{1}{N}\sum_1^N \left(\frac{\|f(m, x_i)\|}{\max_i \|f(m, x_i)\|}\right)^{\infty}$$

Only the term for which $\|f(m, x_i)\|$ is equal to $\max_i \|f(m, x_i)\|$

will survive (=1) after elevation to the ∞ power

→

$\min_m \max_i \|f(m, x_i)\|$: min max

simplified situation:

scalar model : m
scalar data : x

model and data are in the same (scalar) space:

Example: x are noisy measurements of m (ex. thermometer)

$$f(m, x) = m - x$$
(measurement error)

$$k = 2$$

$$\min_m J = \frac{1}{N}\sum_1^N \|f(m, x_i)\|^2$$

$$\rightarrow \min_m J = \frac{1}{N}\sum_1^N (m - x_i)^2$$

$$\frac{\partial J}{\partial m} = 0$$

$$\rightarrow \frac{2}{N}\sum_1^N (m - x_i) = 0$$

$$\rightarrow m = \frac{1}{N}\sum_1^N x_i \quad \textbf{average}$$

sum of displacements = 0

$$k = 1$$

$$\min_m J = \frac{1}{N} \sum_1^N \|f(m, x_i)\|$$

$$\rightarrow \min_m J = \frac{1}{N} \sum_1^N \|m - x_i\|$$

$$\frac{\partial J}{\partial m} = 0$$

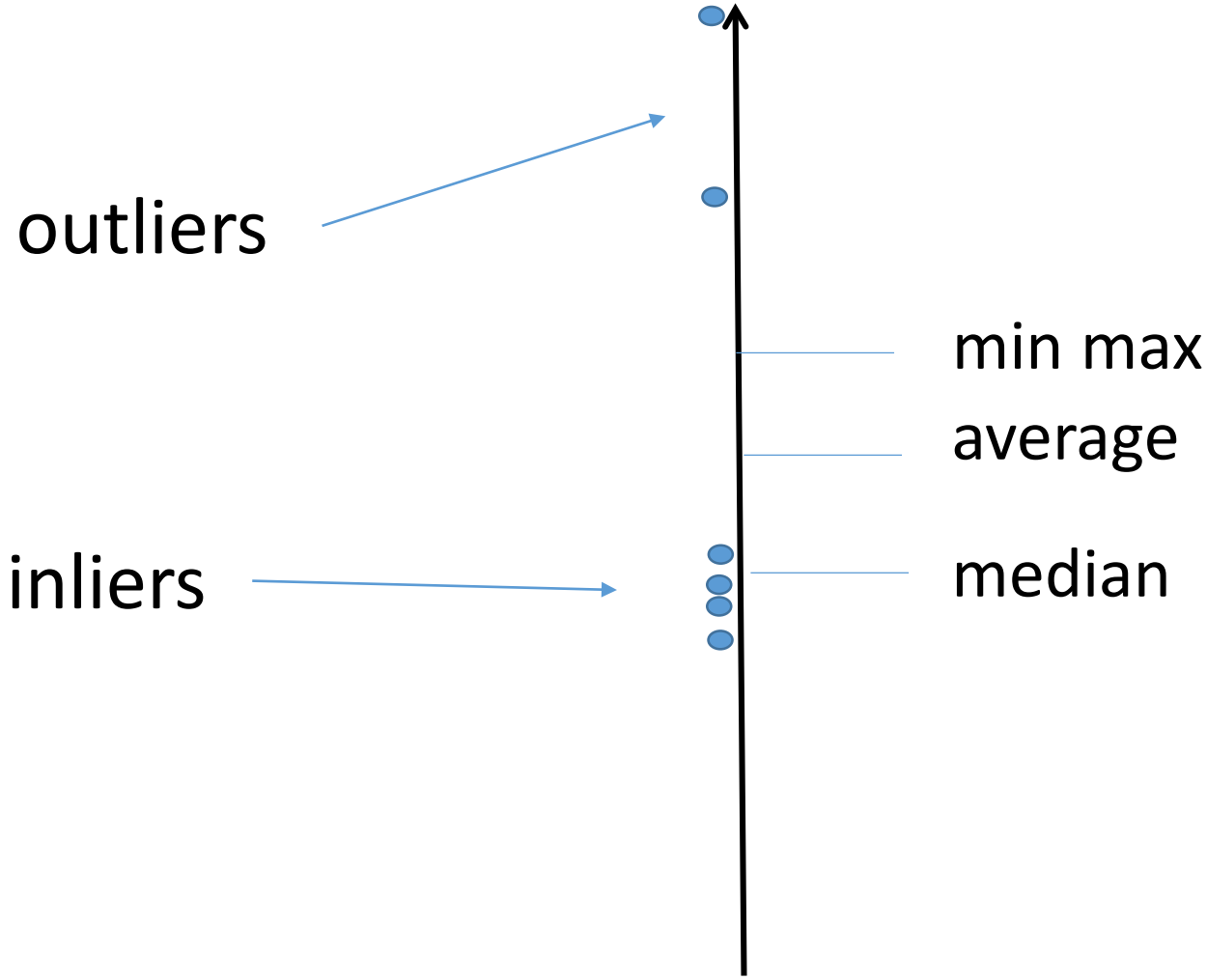
$$\rightarrow \frac{1}{N} \sum_1^N sgn(m - x_i) = 0$$

→ #{data > m} = #{data < m}

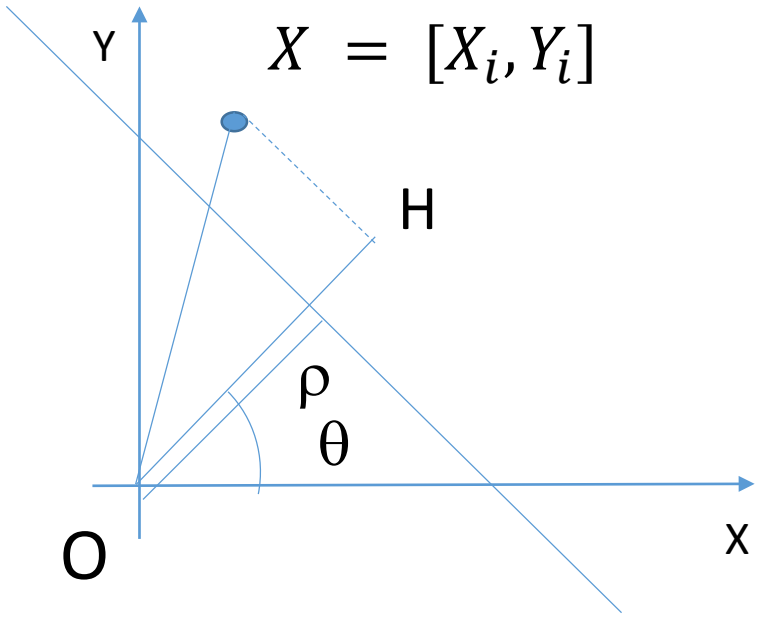
→ **m = Median({x_1 … x_N })**

average and min max: not robust against outliers

median: robust to (< 50%) outliers



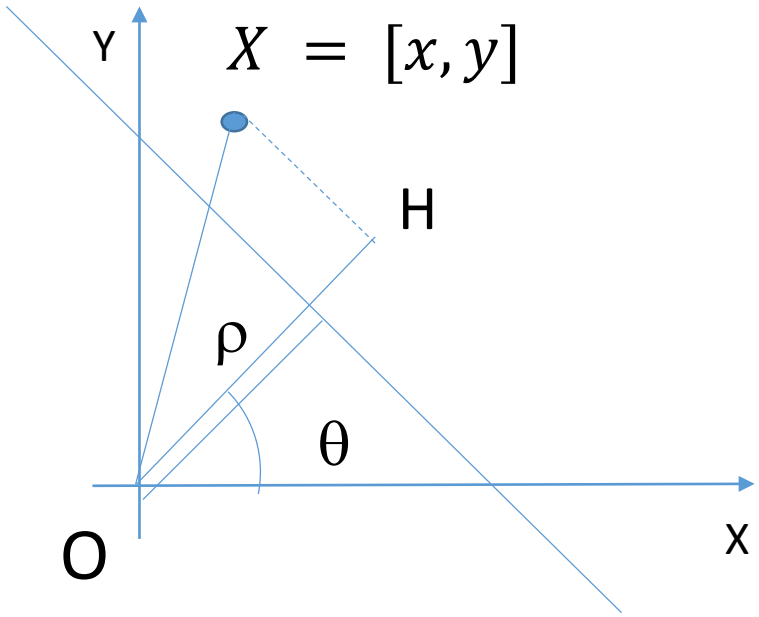
Least squares fitting of **lines** (noisy data points, no outliers)



$$f(m, x_i) = \text{distance (point, line)}$$

$$f(m, x_i) = X_i \cos\vartheta + Y_i \sin\vartheta - \rho$$

Least squares fitting of lines (noisy data points, no outliers)



$$f(m, x_i) = \text{distance (point, line)}$$

$$f(m, x_i) = X_i \cos\vartheta + Y_i \sin\vartheta - \rho$$

a b c

$$\min_{a^2+b^2=1} J = \frac{1}{N}\sum_{i=1}^{N}(aX_i + bY_i + c)^2$$

there are no constraints on c: → $\frac{\partial J}{\partial c} = 0$ → $c = -a\bar{x} - b\bar{y}$

average

$$\min_{a^2+b^2=1} \begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} \overline{x^2} & \overline{xy} & \bar{x} \\ \overline{xy} & \overline{y^2} & \bar{y} \\ \bar{x} & \bar{y} & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

but since $c = -a\bar{x} - b\bar{y}$

the problem is equivalent to

$$\min_{a^2+b^2=1} \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} \overline{x^2} - \bar{x}^2 & \overline{xy} - \bar{x}\bar{y} \\ \overline{xy} - \bar{x}\bar{y} & \overline{y^2} - \bar{y}^2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

A

this is a homogeneous least squares problem

→

solution:

eigenvector associated to the smallest eigenvalue

Since the matrix A is symmetric,

$\begin{bmatrix} a \\ b \end{bmatrix} = U_2$ second column of matrix U produced by

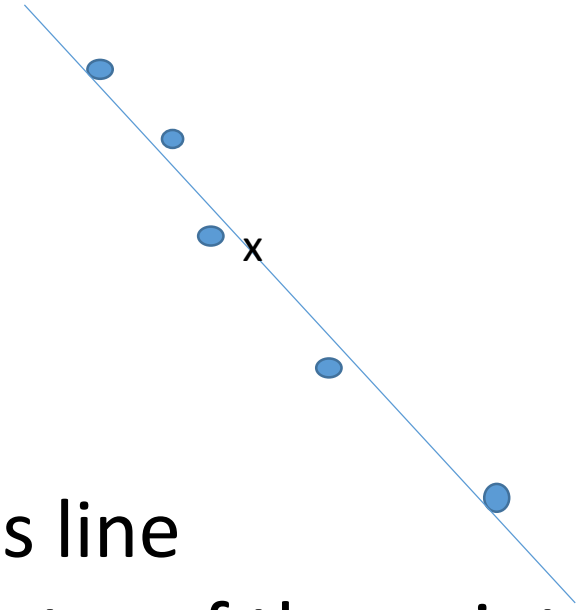
$$\text{svd(A)} = U \begin{bmatrix} e & 0 \\ 0 & f \end{bmatrix} U^T$$

Least squares line

the signed distances

$$f(m, x_i) = X_i \cos\vartheta + Y_i \sin\vartheta - \rho$$

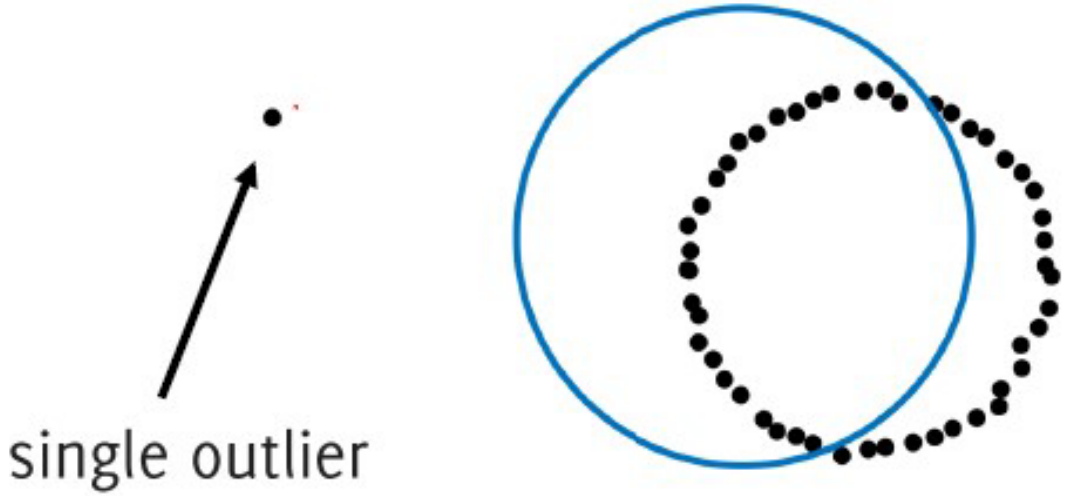
sum up to 0

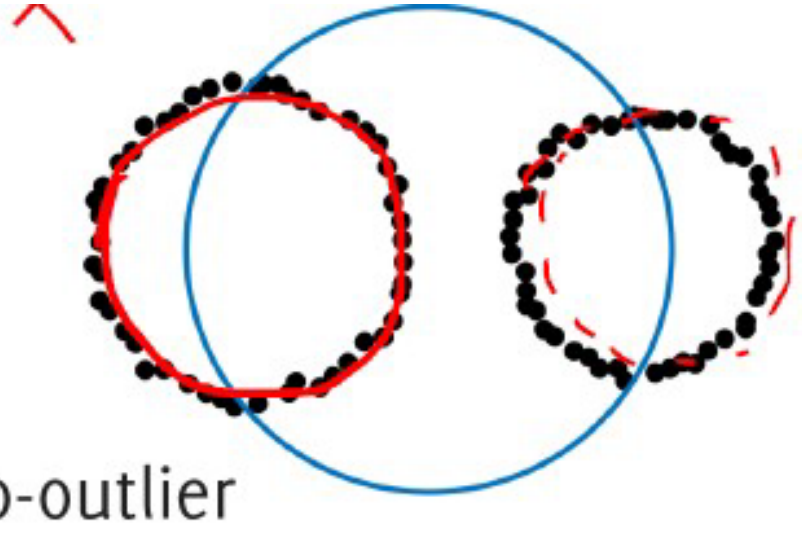


Least squares line

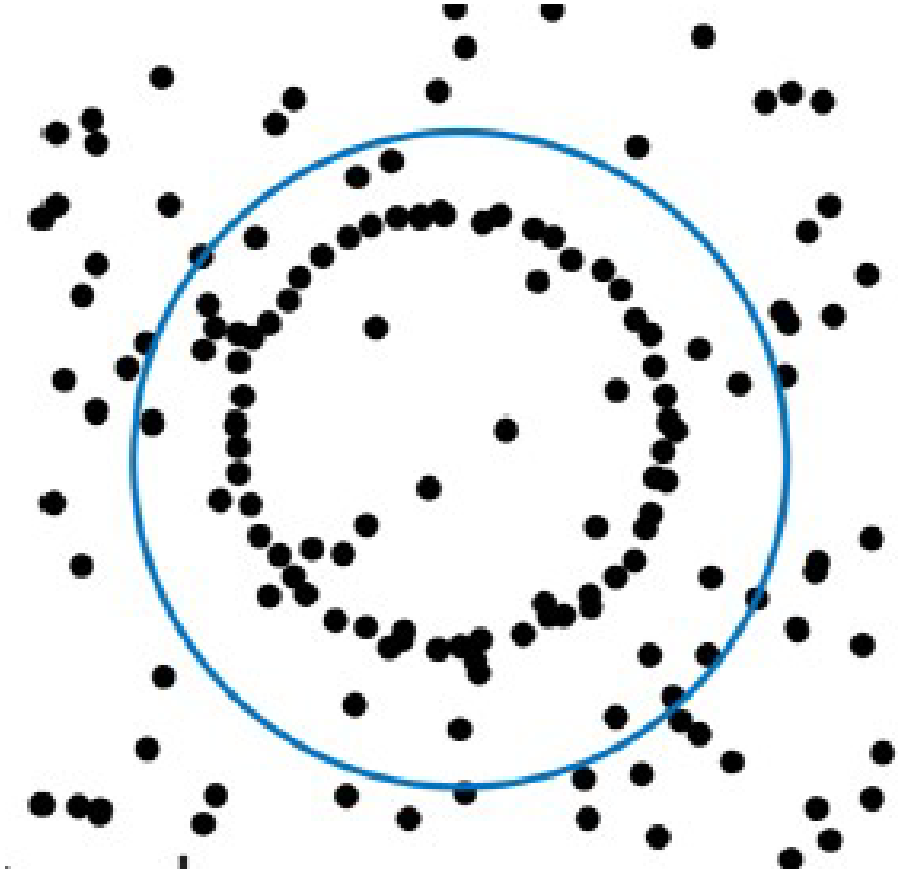
goes through the geo-center of the points

Model fitting to noisy images: OUTLIERS



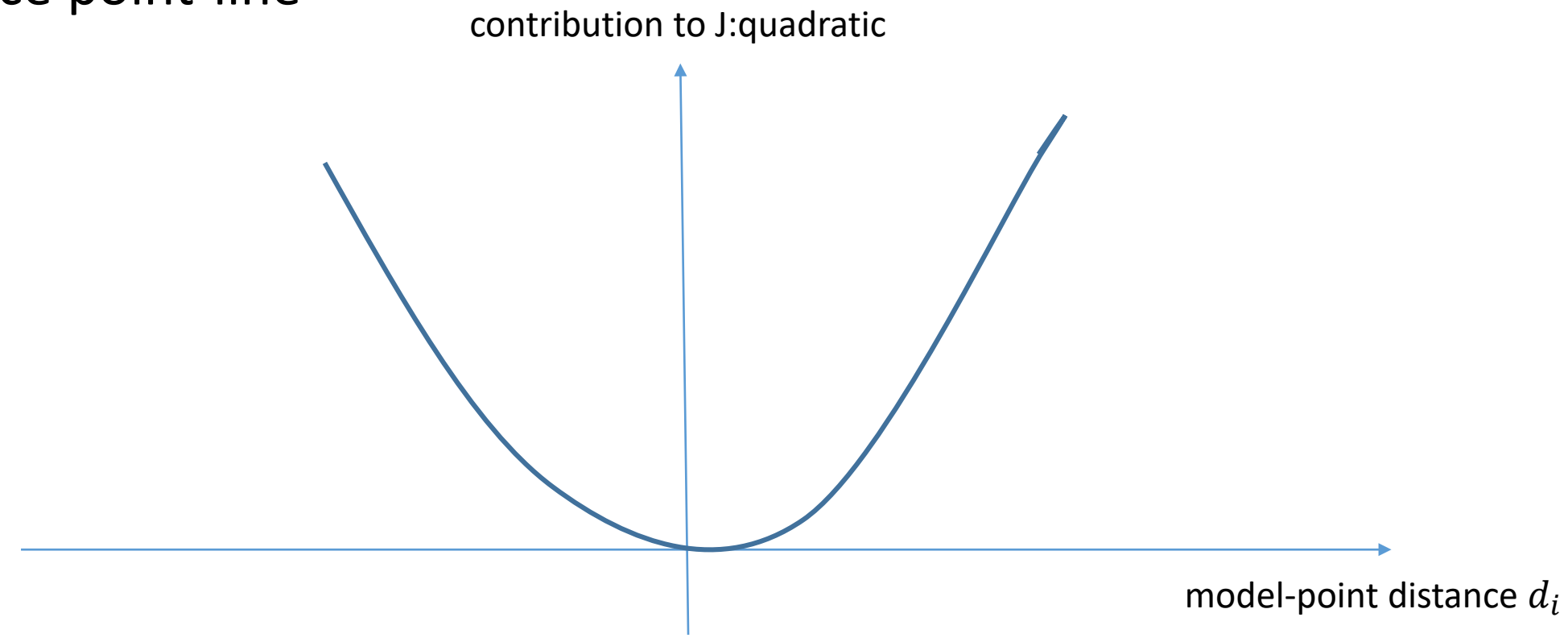


unstructured outliers



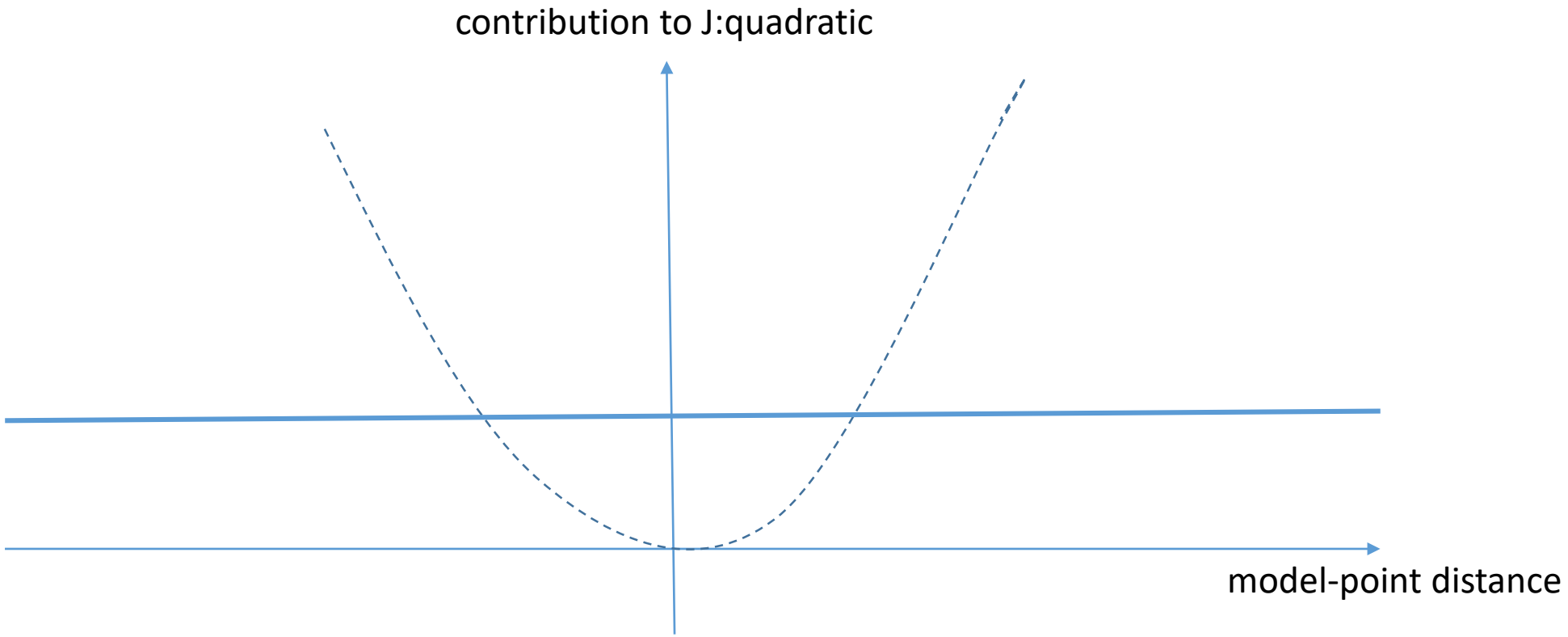
Robust estimation

- **Inlier** (datum x_i **close** to model): J approximately quadratic with the distance point-line



Robust estimation

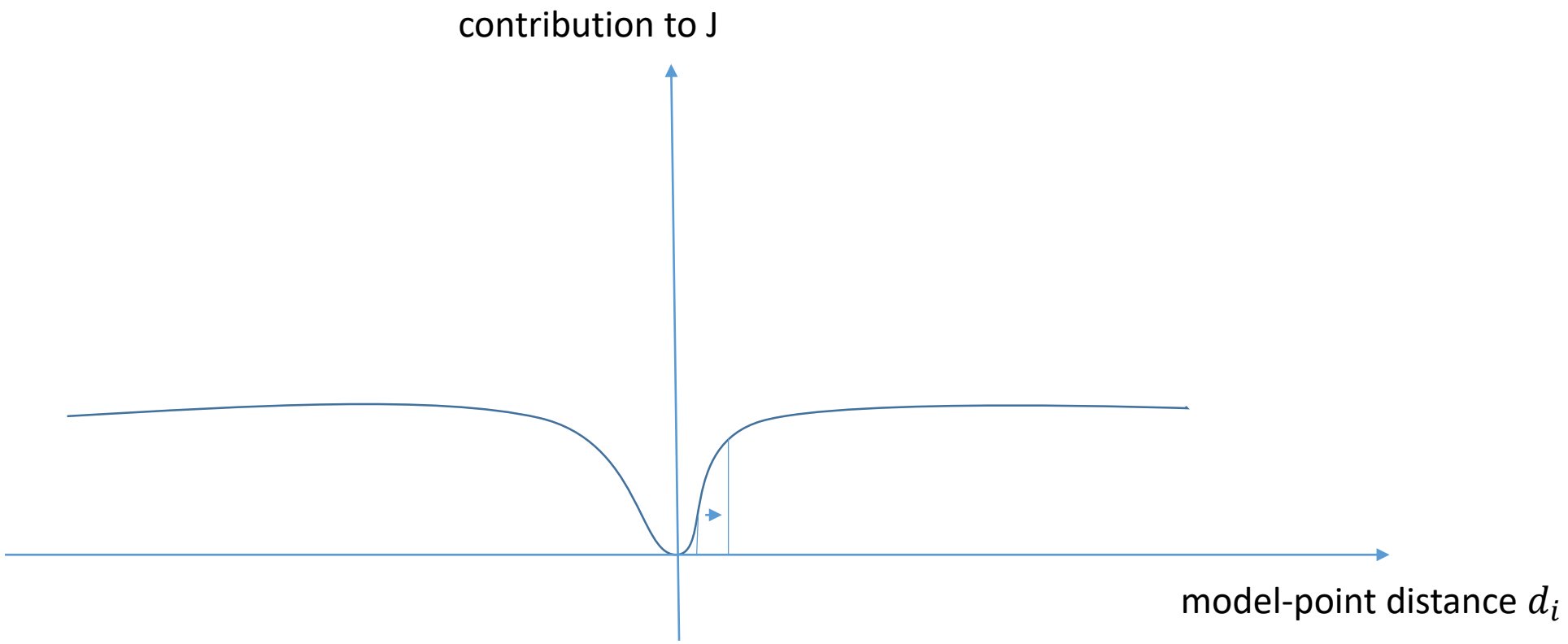
- **Outlier** (datum x_i **far** from the model): J approximately constant with the distance point-line → indifferent → does not affect estimate



Contribution to J through a datum d_i close to the model

- quadratic close to the model $\frac{\partial J}{\partial m} = \sum \frac{\partial J}{\partial d_i} \frac{\partial d_i}{\partial m} = \sum d_i \frac{\partial d_i}{\partial m}$

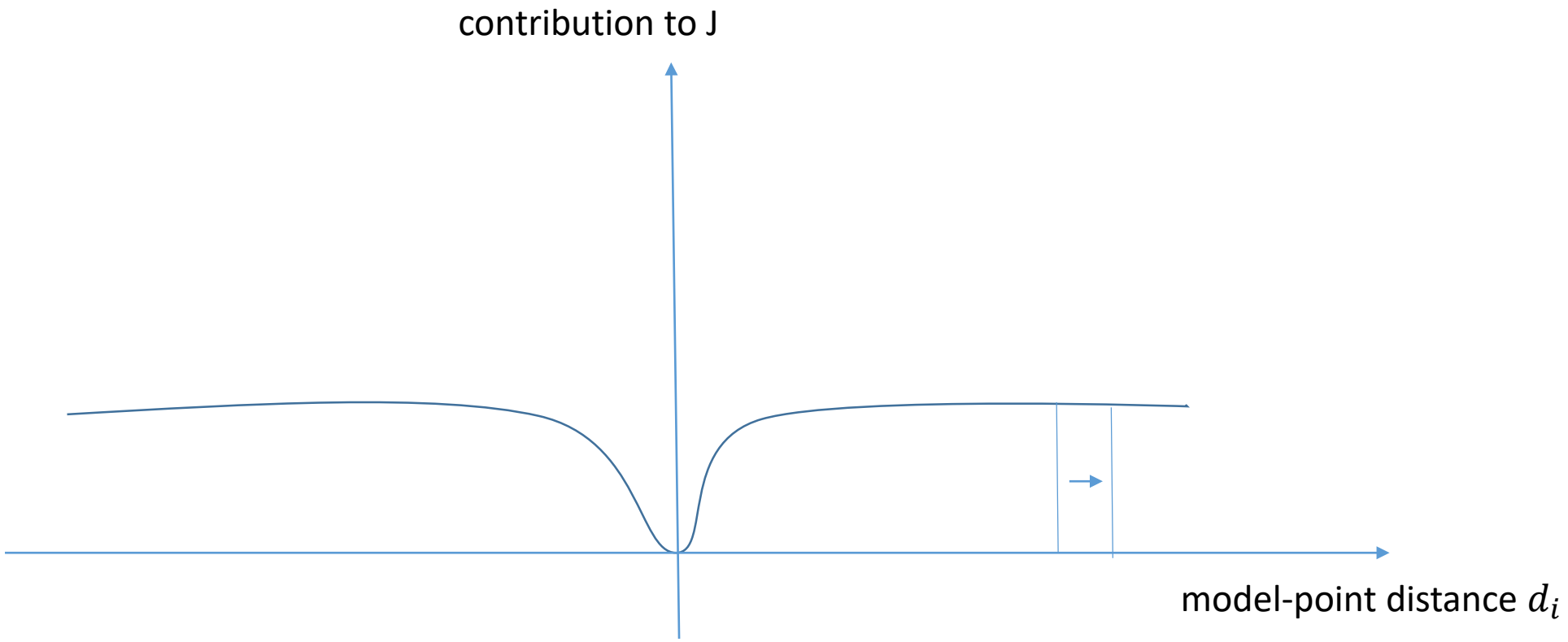
- …



Contribution to J through a datum d_i far from the model

- quadratic close to the model
- constant far from the model

→ disappears when differentiating $\frac{\partial J}{\partial m} = \frac{\partial J}{\partial d_i}\frac{\partial d_i}{\partial m} = 0$



Robust estimation

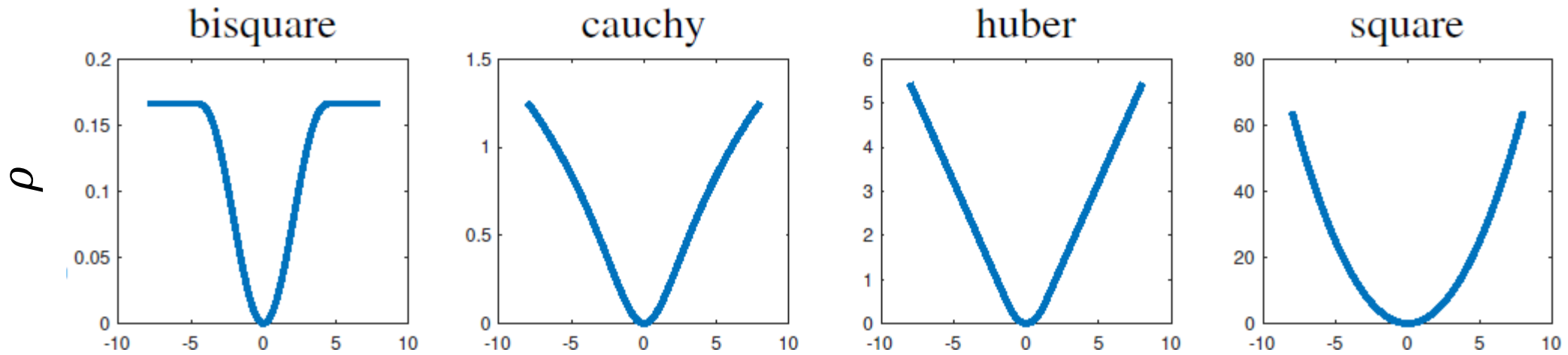
a possible function

$$\arg\min_m \frac{1}{N}\sum_{i=1}^{N}\frac{\|f(m,x_i)\|^2}{\sigma^2+\|f(m,x_i)\|^2}$$

- degree N
- many local extrema
- → difficult to minimize

M-Estimator

- Temporary change of notation: $d_i = f(m, x_i) \rightarrow r_i$ and $m \rightarrow \theta$
- Replaces the squared loss with a different loss function which penalizes less large residual values (deemed to correspond to outliers)
- $\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \sum_{i=1}^{N} \rho(r_i(\theta))$
- Where ρ a symmetric, positive-definite function having a unique minimum in zero



- Andrea Fusiello, Elaborazione delle Immagini: Visione Computazionale,

M-Estimator

- To solve this minimization problem we need to zero the derivative for each dimension of θ
- $\sum_{i=1}^{N} \rho'(r_i) \frac{\partial r_i}{\partial \theta_j} = 0, \quad j = 1, \dots, M$
- The trick is to associate to ρ a weight function w such that

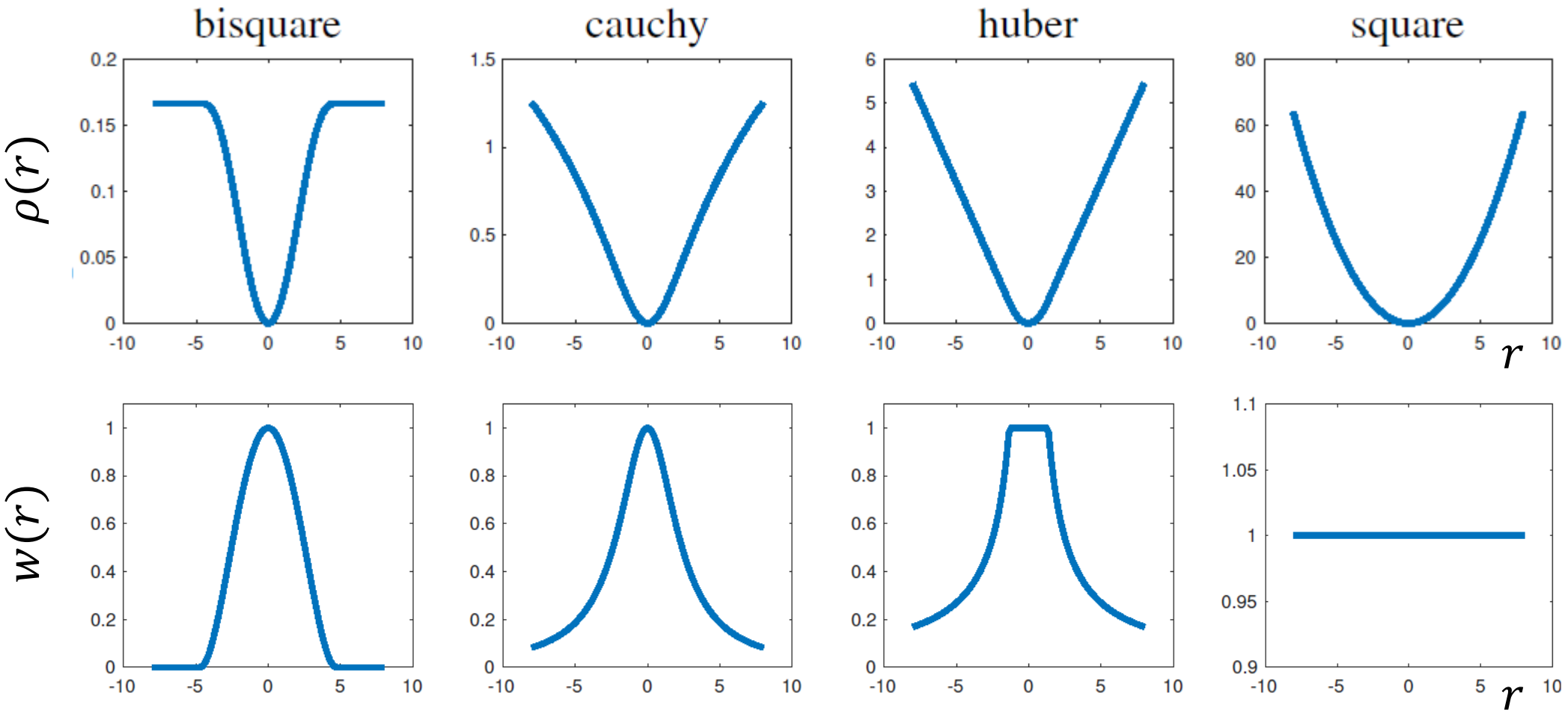
$$\rho'(x) = x * w(x)$$

$$\rightarrow$$

$$\rho'(r_i) = r_i * w(r_i)$$

M-Estimator

- Weights associated to the previous losses are



- Andrea Fusiello, Elaborazione delle Immagini: Visione Computazionale

M-Estimator

- Then the zeroing the derivative corresponds to solving the following
- $\sum_{i=1}^{N} r_i \ * w(r_i) \frac{\partial r_i}{\partial \theta_j} = 0 \qquad j = 1, \dots, m \quad (1)$
- If we consider an iterative scheme where at iteration k, we alternate weight definition and minimization for r_i. Then, during the $k - th$ iteration we treat weights as fixed and defined by $r_i^{(k-1)}$, such that we solve (1) just w.r.t. to r_i. In this case, $\hat{\theta}$ is the same solution of
- $\underset{\theta}{\operatorname{argmin}} \sum_{i=1}^{N} w\left(r_i^{(k-1)}\right) r_i^2$
- Which is a weighted least square problem!

M-Estimator

- Minimizing this
- $\hat{\theta}^{(k)} = \underset{\theta}{\operatorname{argmin}} \sum_i \; w(r_i^{(k-1)})\, r_i(\theta)^2$
- Can be done by weighted least square, but weights w depend on $r_i(\theta)$
- Iterative Reweighted Least Sqares (IRLS) alternates
- Define weights $w(r_i^{(k-1)})$ from residual $r_i\left(\hat{\theta}^{(k-1)}\right)$ at the previous iteration. This is done by simply sampling the weight curves.
- Define the model $\hat{\theta}^{(k)}$ by solving the weighted least square keeping $w(r_i^{(k-1)})$ as fixed

- Andrea Fusiello, Elaborazione delle Immagini: Visione Computazionale,

The Weights and ρ

- What actually determines the M-estimator is the influence function ρ'.
- ρ' in robust estimator should be **bounded**. For example r^2 has not a bounded derivative. The others shown before have a bounded derivative since $\rho'(x) = w(x) * x$ decay faster than linearly.
- The function ρ should be strictly convex, to yield a unique minimum

- Obviously, the weights should decrease as r increases, to bound residual contribution.

OUTLIERS

Basic idea: VOTING APPROACH

Data are unlabelled

→

- a datum x_i votes for a model m if they are compatible $f(m, x_i) \cong 0$

- most voted models m are selected as model estimate(s)

Main approaches

- HOUGH TRANSFORM (1961 Duda – Hart)
- RANdom SAmple Consensus «RANSAC» (1980 Fischler – Bolles)
- Variations of RANSAC

HOUGH TRANSFORM

Each individual datum $x_i \in X$ votes for all the models m that are compatible with it:

$$HT(x_i) = \{m \in \Re^k : f(m, x_i) = 0\}$$

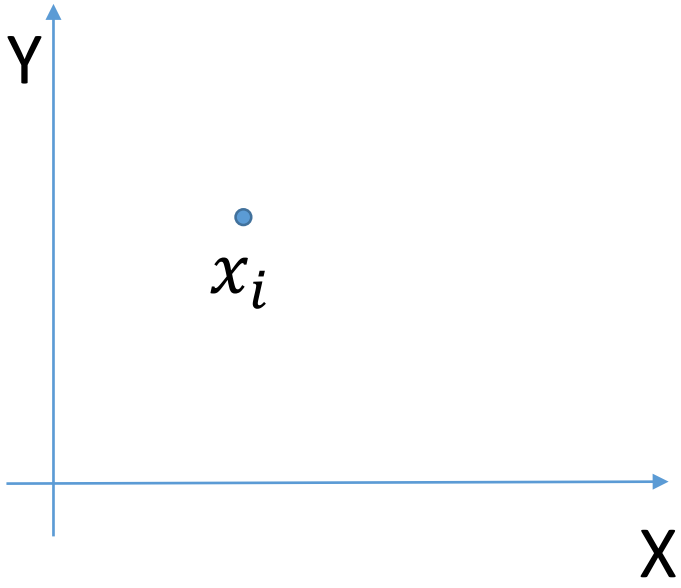
The number of votes obtained by m is

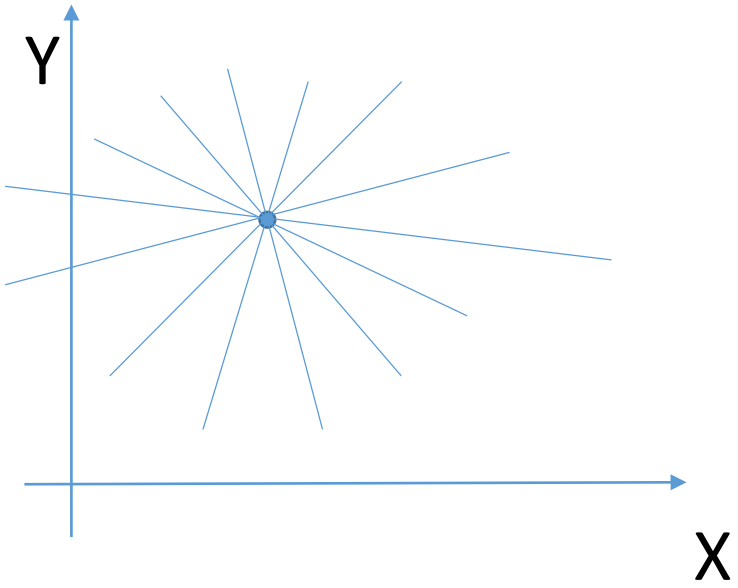
$$V_X(m) = \#\{x_i \in X : f(m, x_i) = 0\}$$

Relative maxima of $V_X(m)$ are selected as

$$\arg rel \max_m V_X(m) = \#\{x_i \in X : f(m, x_i) = 0\}$$

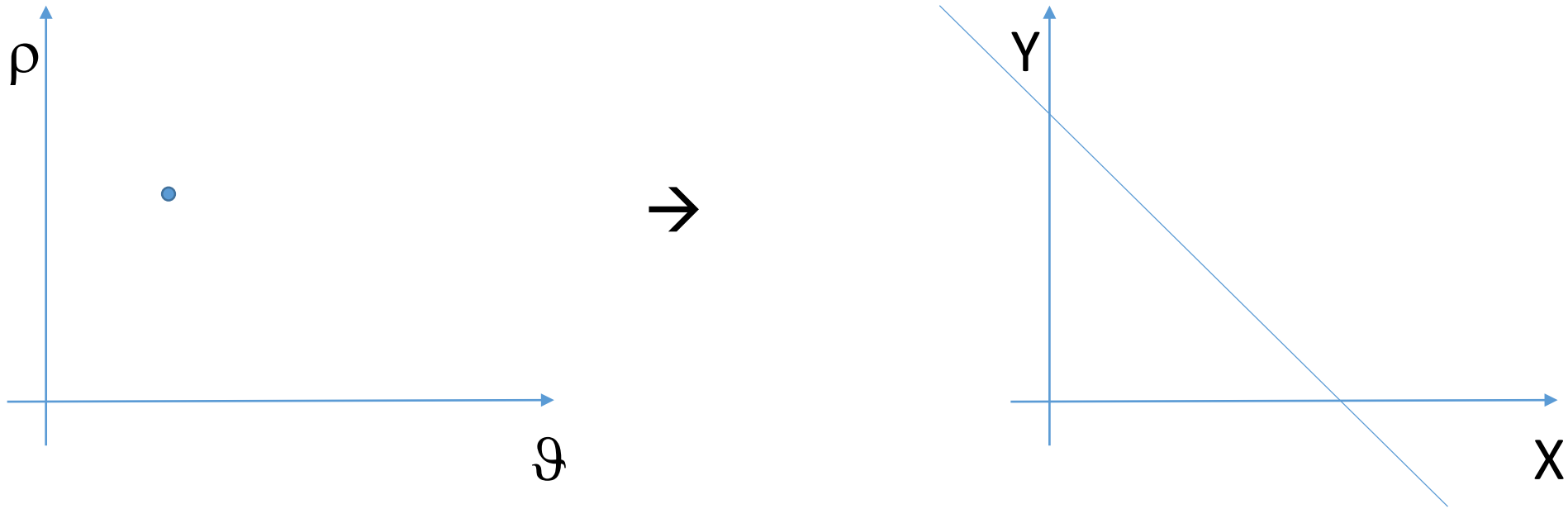
HOUGH TRANSFORM of a datum:
= set of models compatible with it





$$HT(x_i) = \{m \in \Re^k : f(m, x_i) = 0\}$$

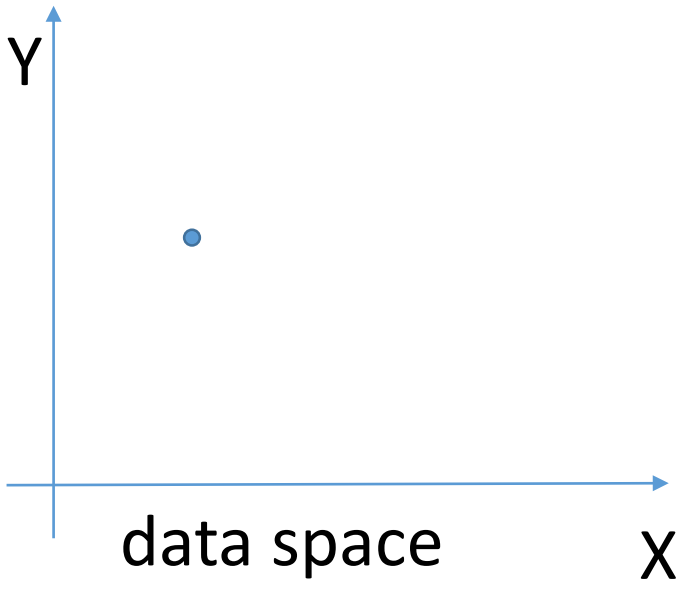
Hough space: the space of model parameters

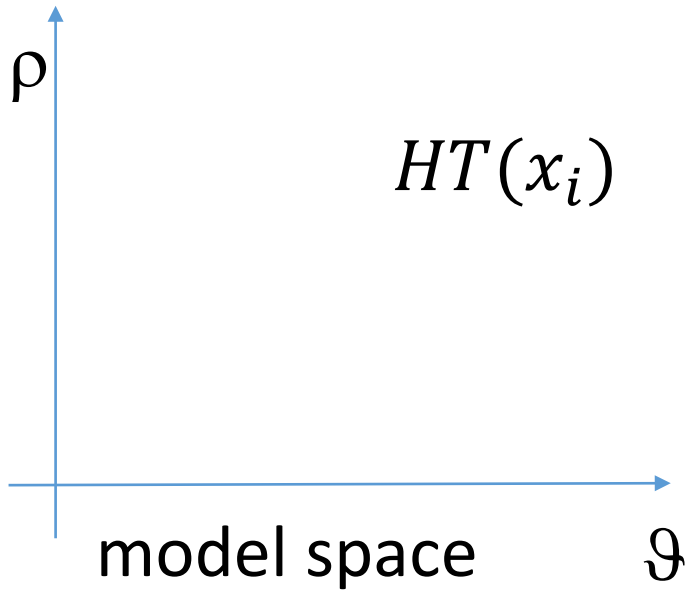


a point in the model space represents a line in the datum (coordinate) space

What is the representation of a datum in the model space?

- Set of models (e.g., lines) compatible with the datum



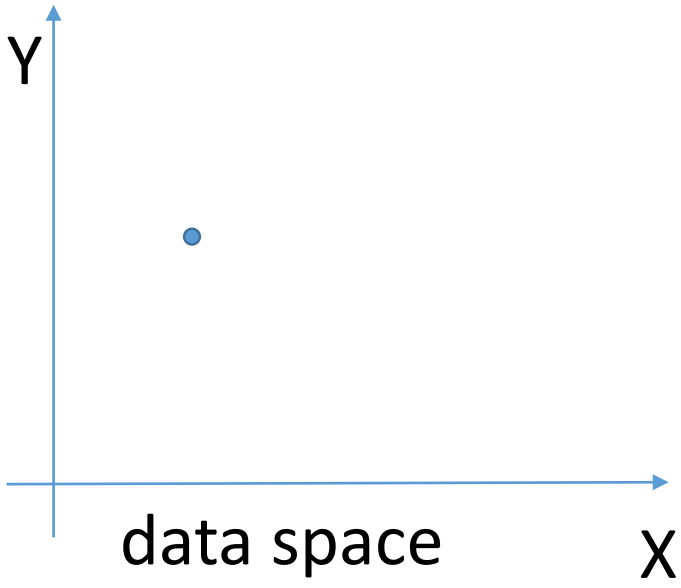


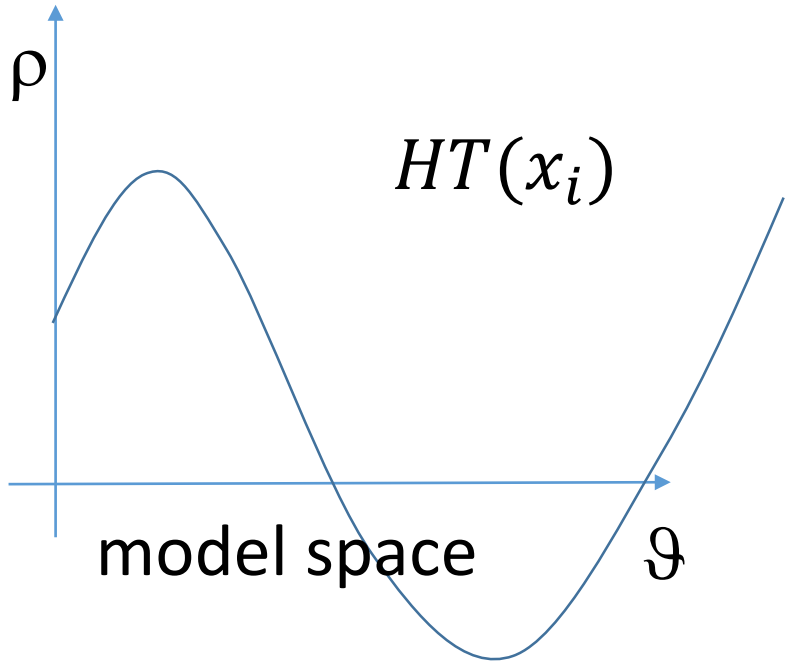
answer based on equation $f(m, x_i) = X_i \cos\vartheta + Y_i \sin\vartheta - \rho = 0$

$$X_i \cos\vartheta + Y_i \sin\vartheta = \rho$$

Hough Transform of a datum (e.g., a feature point):

- Set of models (e.g., lines) compatible with the datum





$$f(m, x_i) = X_i \cos\vartheta + Y_i \sin\vartheta - \rho = 0$$

Line equation → set of points on the line

$$X_i \cos\vartheta + Y_i \sin\vartheta - \rho = 0$$

given, fixed coordinates

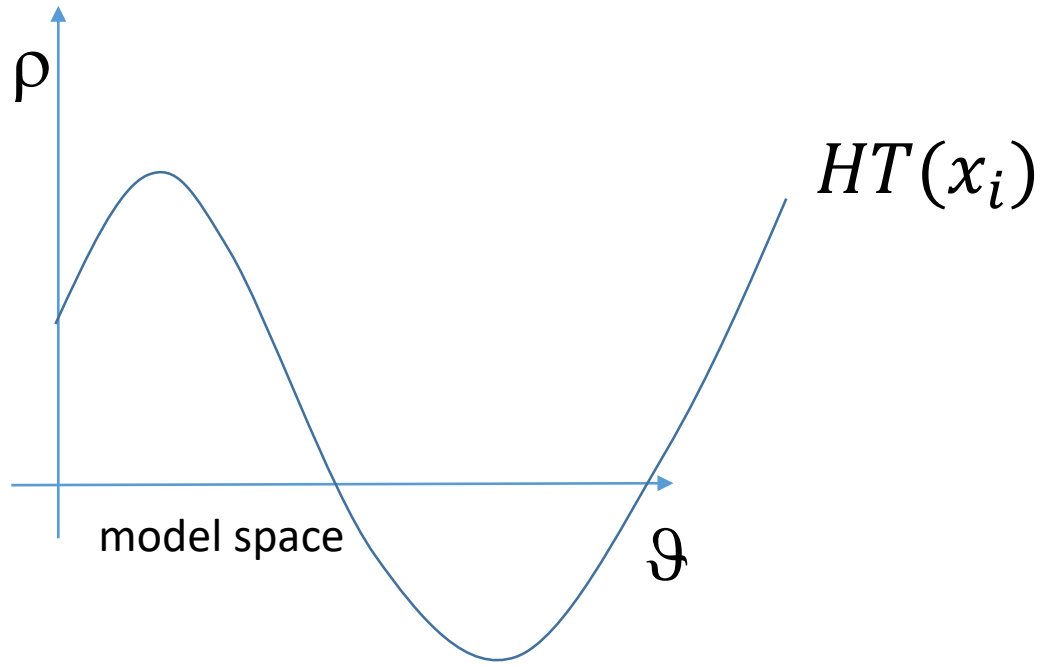
unknown variable parameters

$$\rho = X_i \cos\vartheta + Y_i \sin\vartheta$$

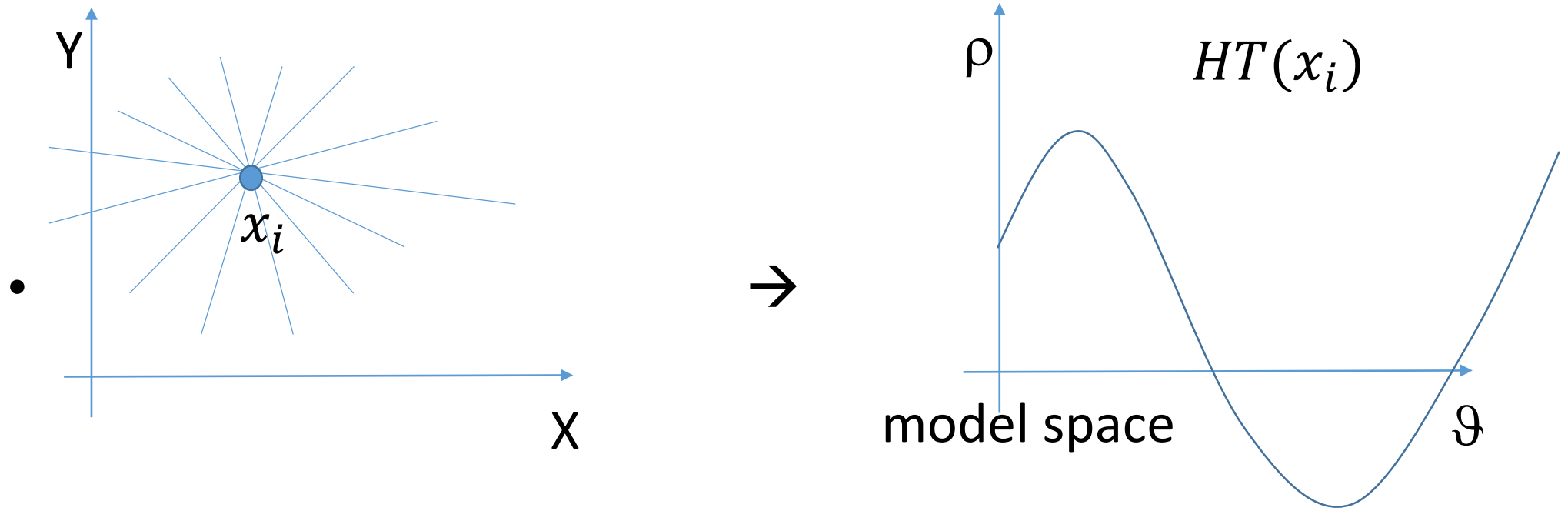
«point» equation → set of lines through the point

HT of a point (as a set of lines): sinusoid

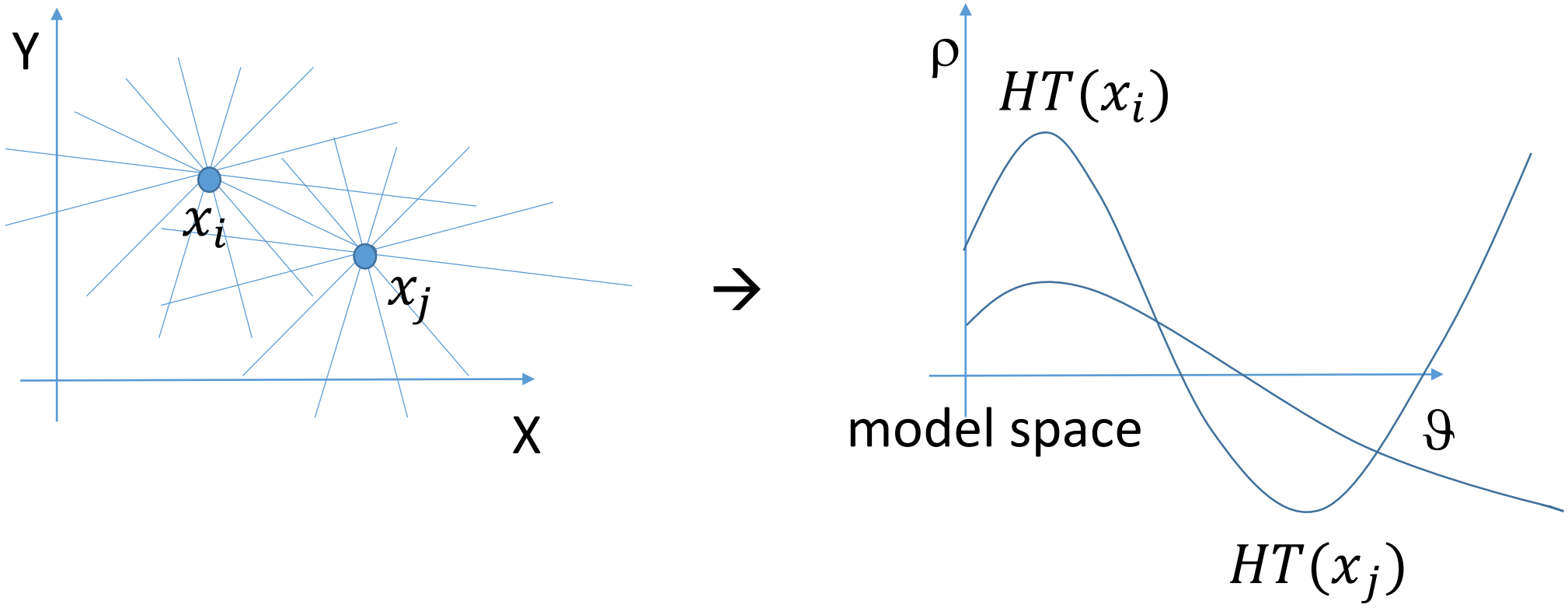
$$\rho = X_i \cos\vartheta + Y_i \sin\vartheta$$



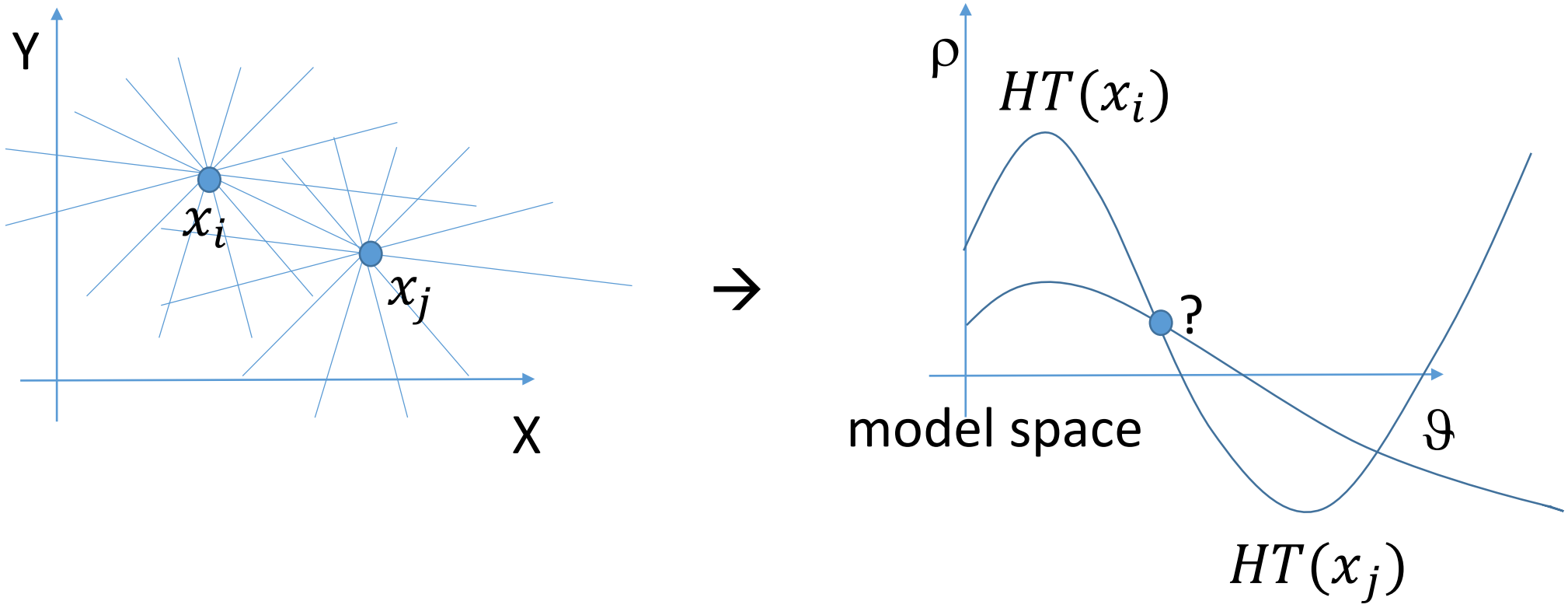
Hough transform of a point:



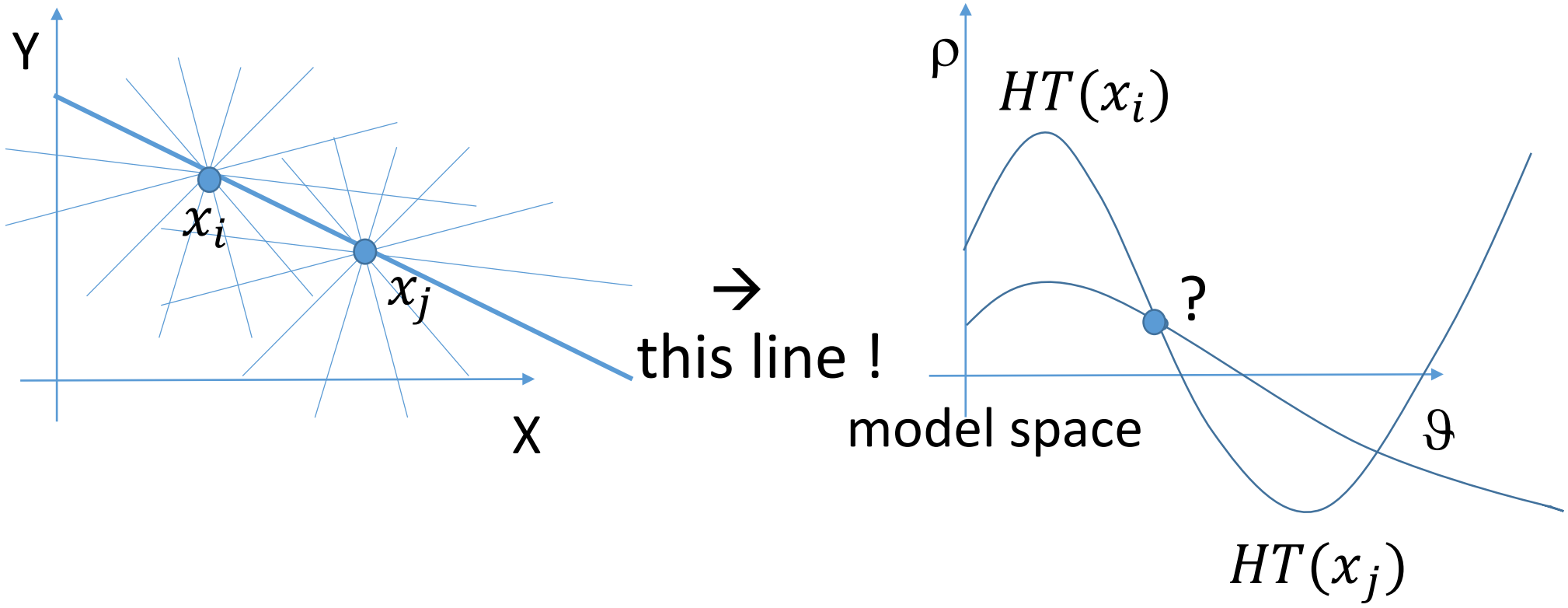
Hough transform of a second point:



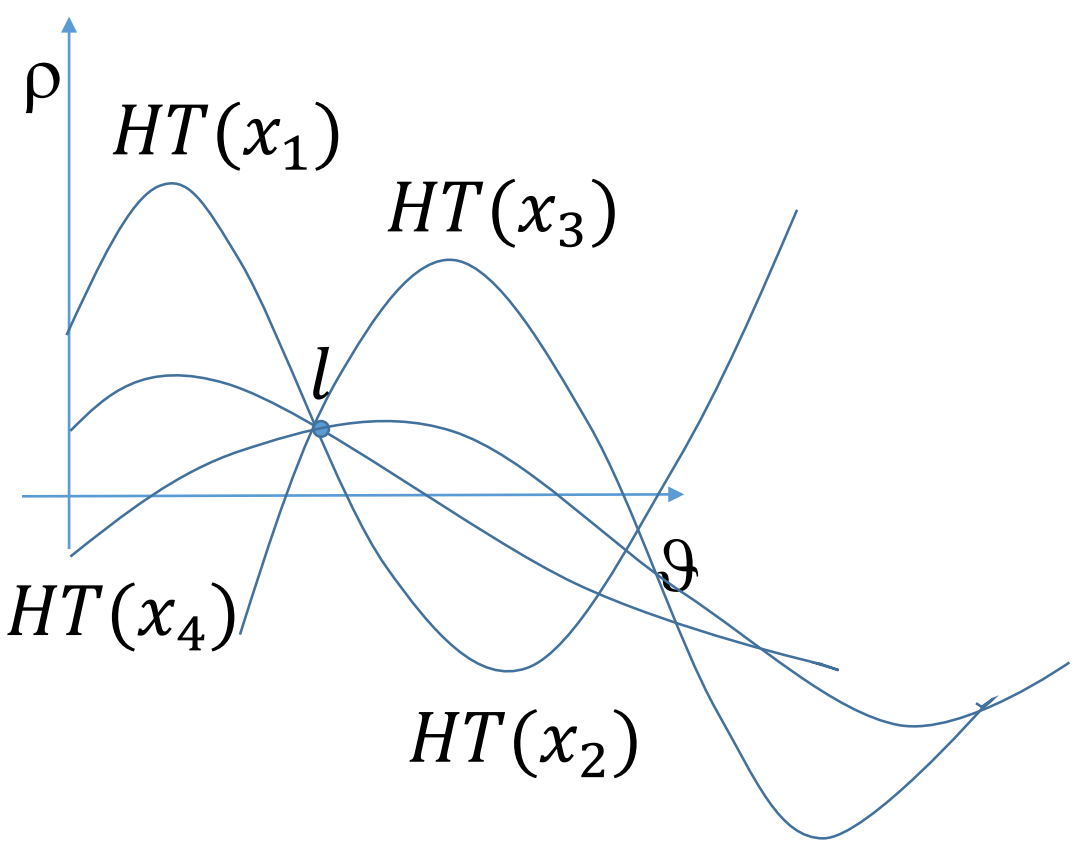
Hough transform of a second point:

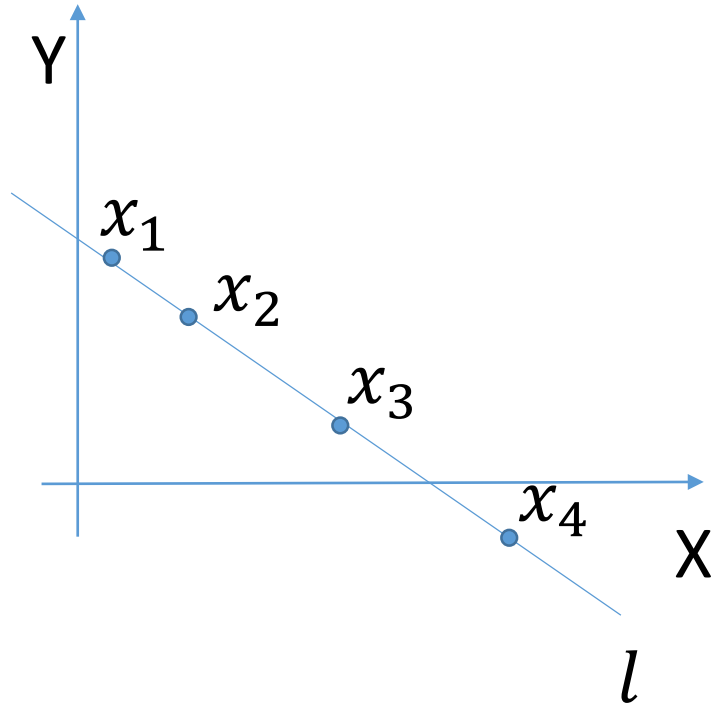


Hough transform of a second point:

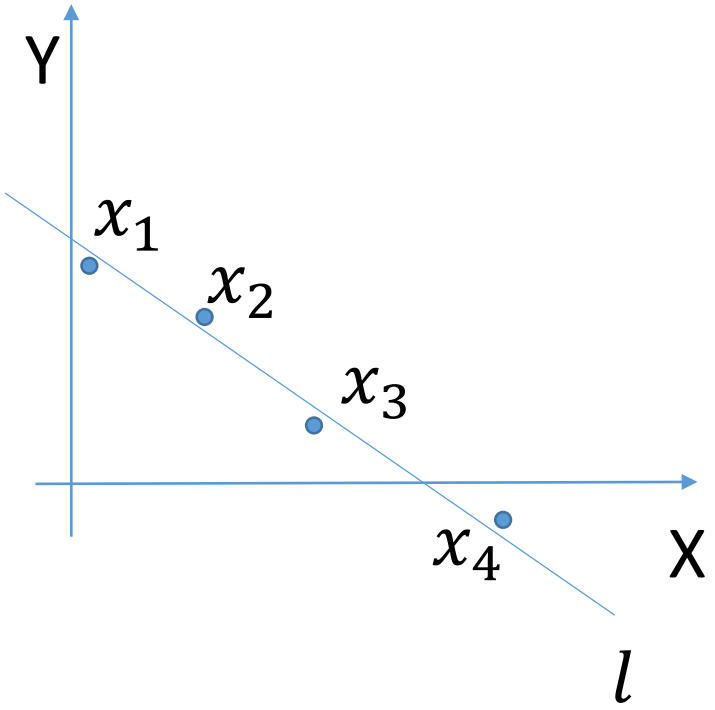


concurrent HTs → colinear points



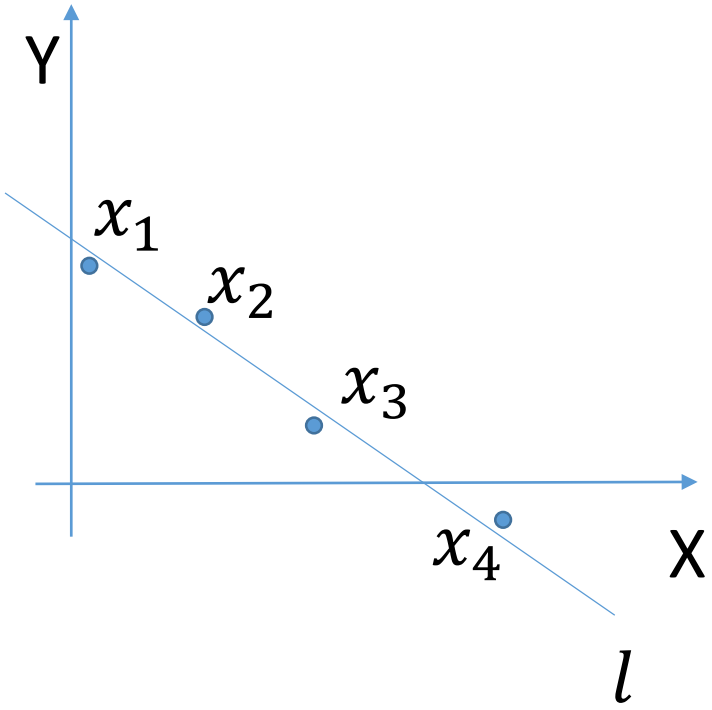


Noise → almost colinear points

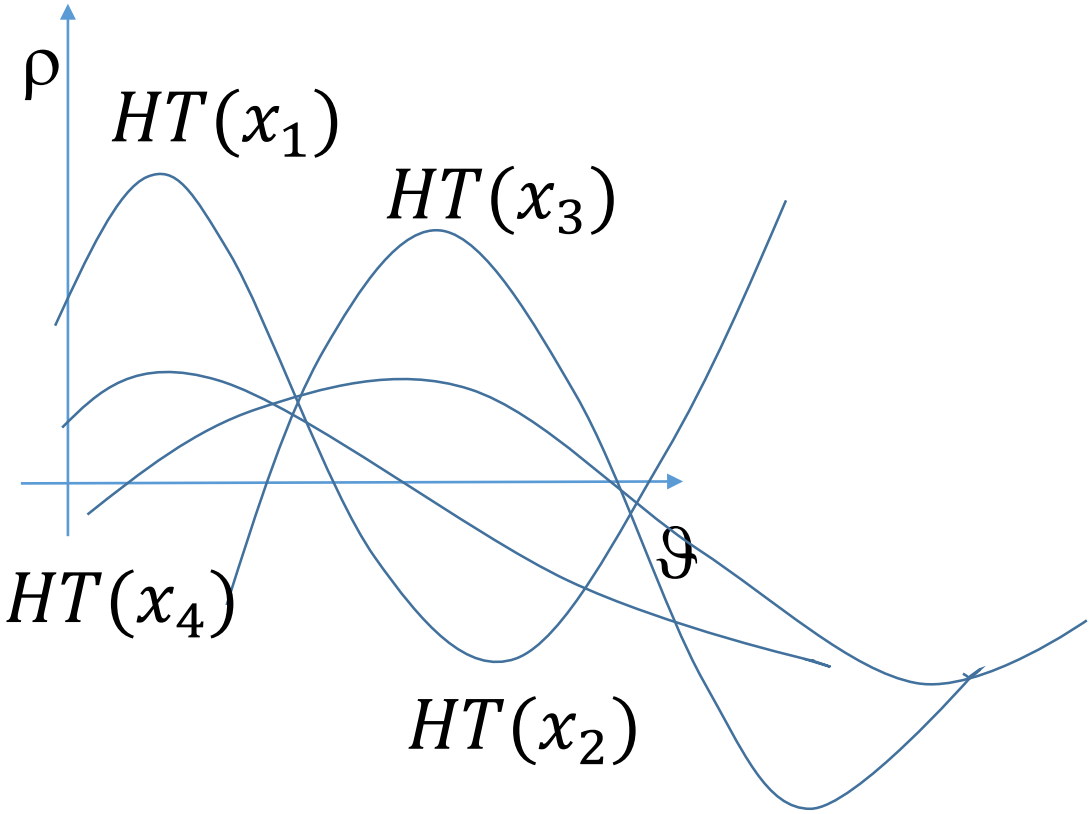


→ ?

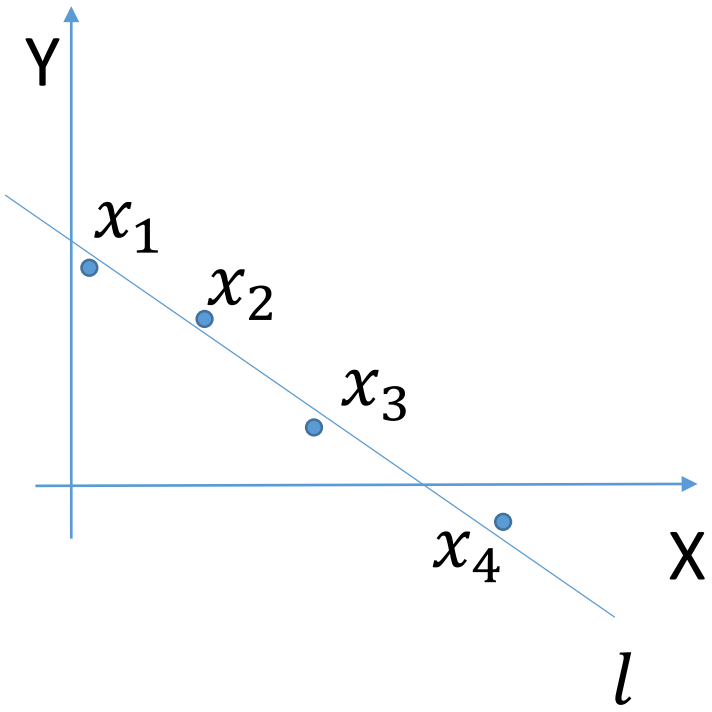
almost colinear points → almost concurrent HTs



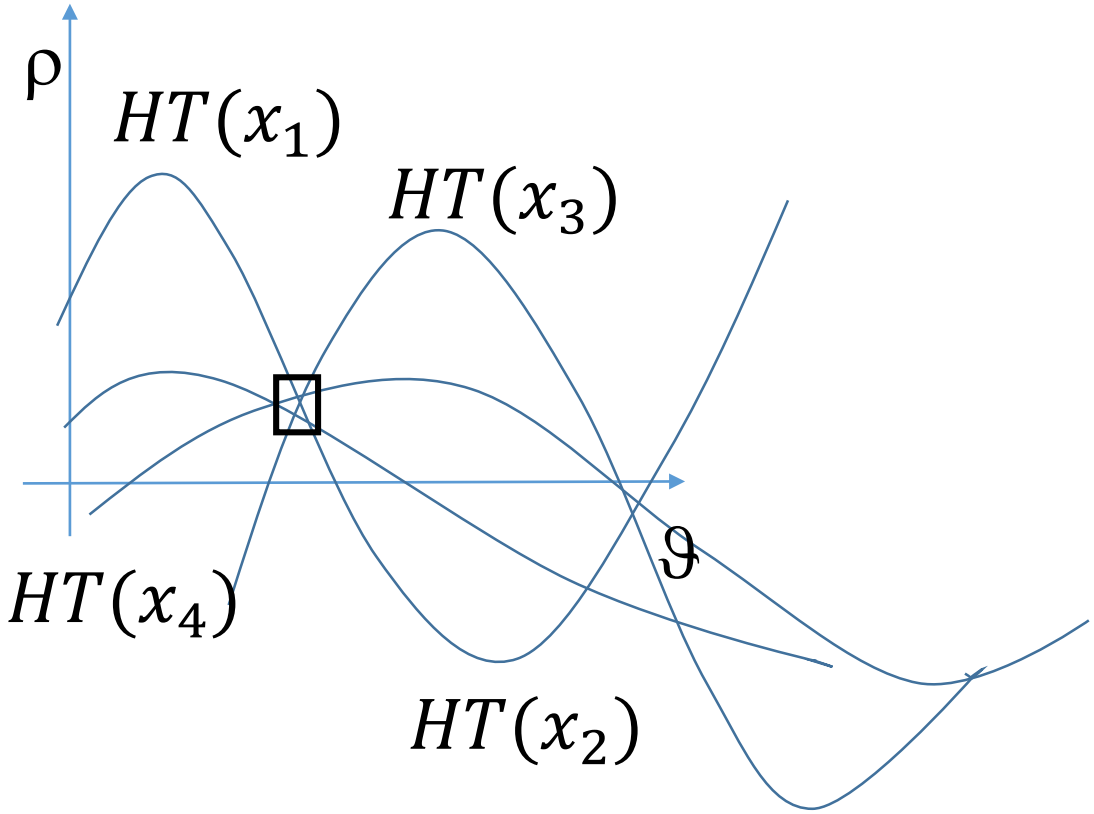
→ ?



almost colinear points → almost concurrent HTs



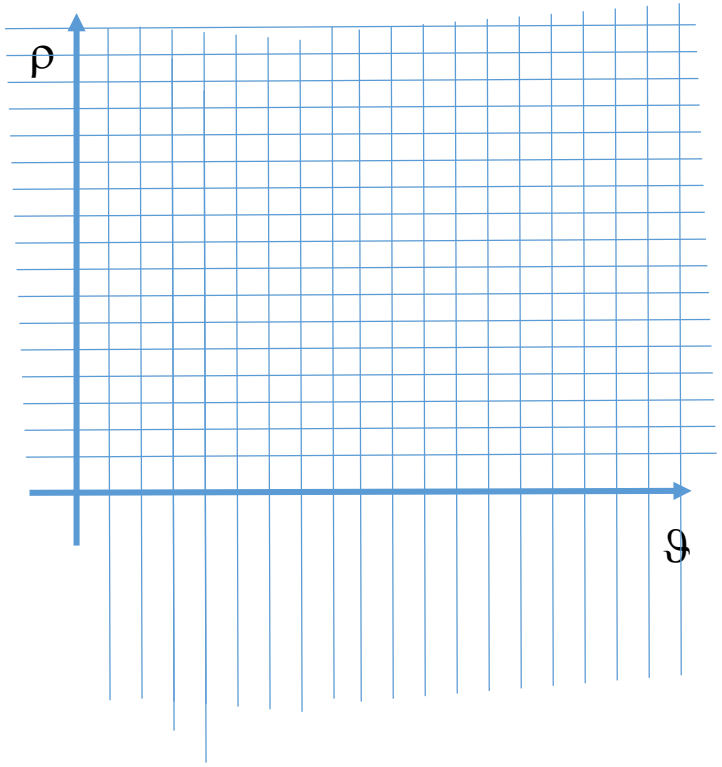
→ ?



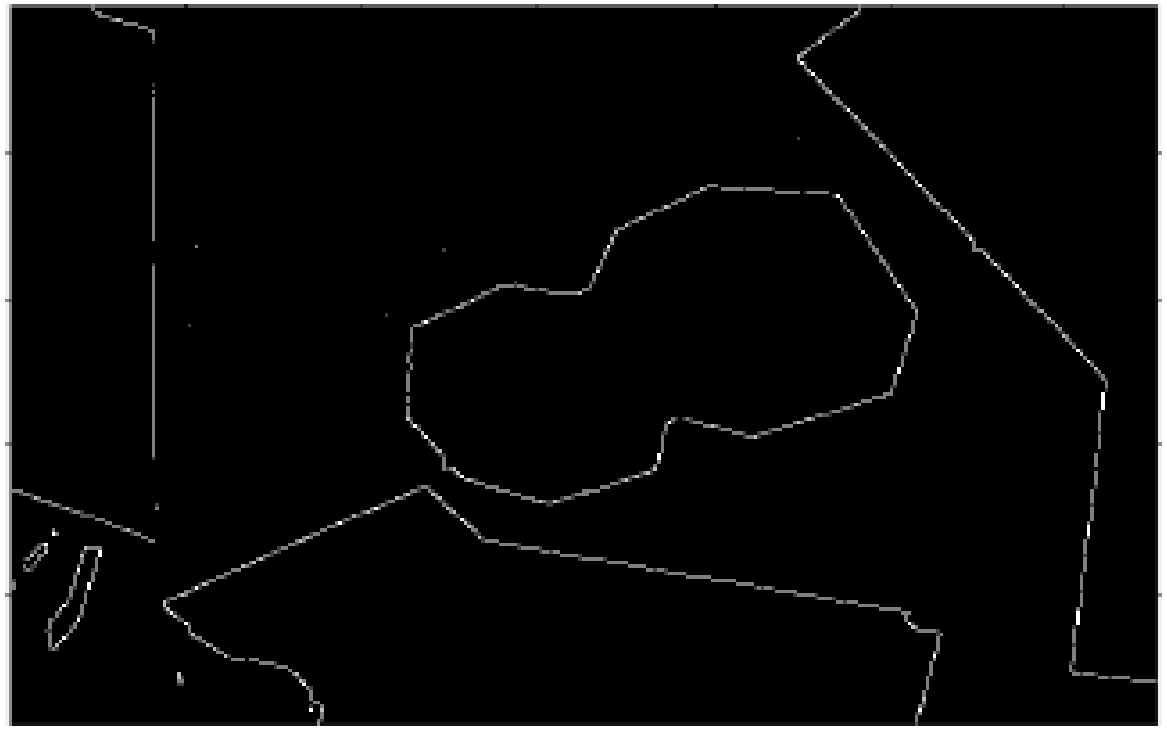
HTs cross a common (small) neighborhood

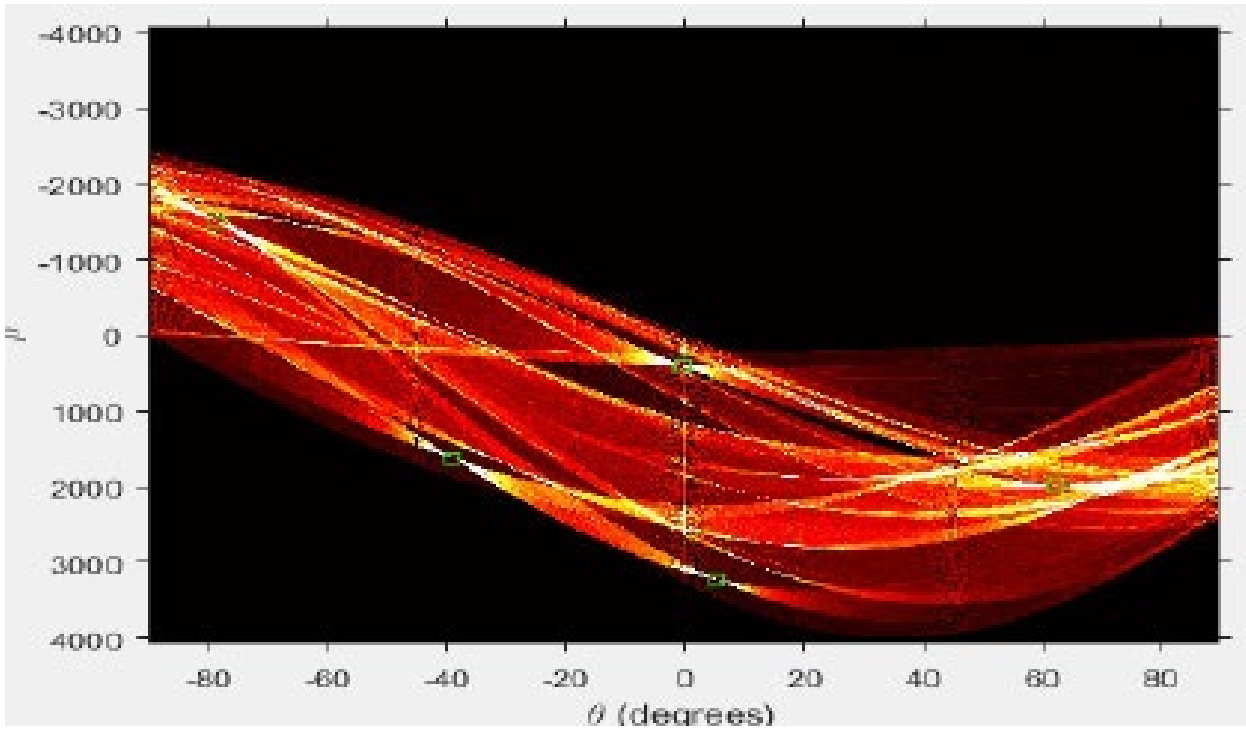
Implementation (finite approximation to HT)

- Discretization of the Hough Space (i.e., the model space) into cells
  - For each cell, set a vote counter = 0
  - For each datum (point)
    - Compute the HT
      - Determine cells crossed by HT
        - Increment vote counter for crossed cells
      - At the end, select cells where vote counter is > threshold and is a local max
    - Refine line estimate by refitting to points that voted for that line



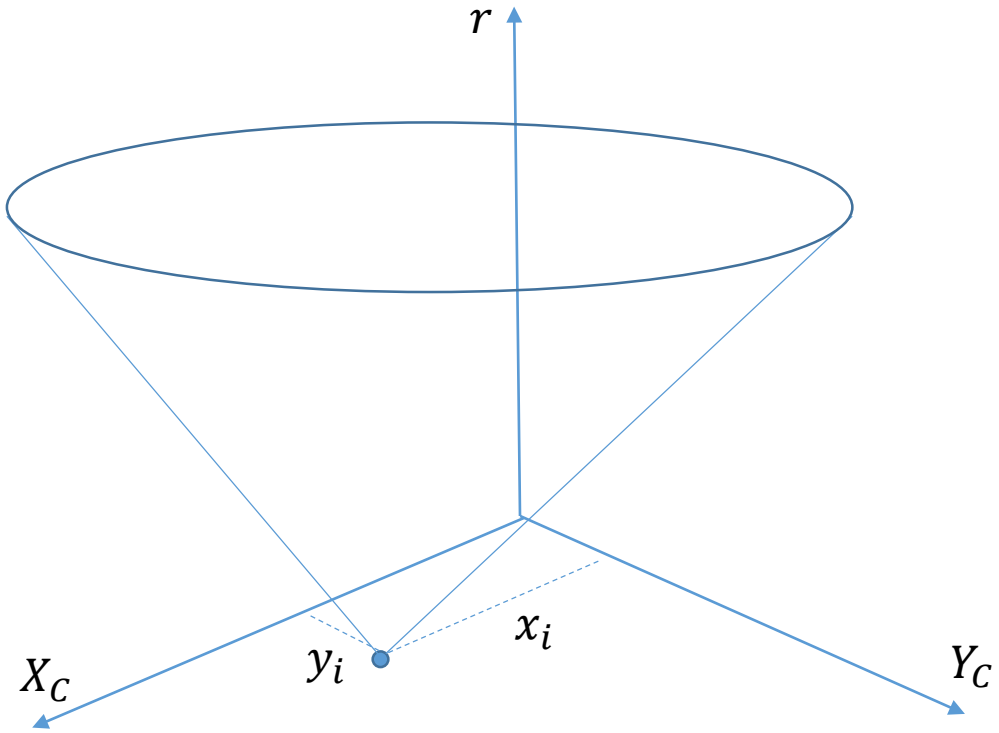
edgepoints, HTs (sinusoids) and local maxima





Question: HT of a point …
in the space of the parameters of a circumference?

$$r^2 = (x_i - X_C)^2 + (y_i - Y_C)^2$$



Hough Transform: computational complexity

- Proportional to the number of cells:

$$O(M^k)$$

where M is the number of subdivisions, and
k is the number of model parameters (d.o.f)

Exponentially grows with the complexity of the model:

• lines:	k = 2	OK
• circumferences:	k = 3	feasible
• conics:	k = 5	not feasible

RANSAC: Random Sample Consensus

- Complex models : conics, homographies
- Random step → solution found *with a certain probability*

- Hough Transform: each datum votes for all models compatible with it
- RANSAC: data vote for **proposed** models

- SAMPLE: a minimal group of data compatible with just one single common model

→

Each SAMPLE **proposes** just ONE model → other data VOTE for it

SAMPLE

MODEL

- line
- circumference
- conic
- homography (plane to plane)
- fundamental matrix

- SAMPLE

- 2 points
- 3 points
- 5 points
- 4 pairs of corresponding points
- 7-8 pairs of corresponding points

$k > 3$: too many samples (e.g. k=3: N(N-1)(N-2)/3!)

→ not all samples are considered

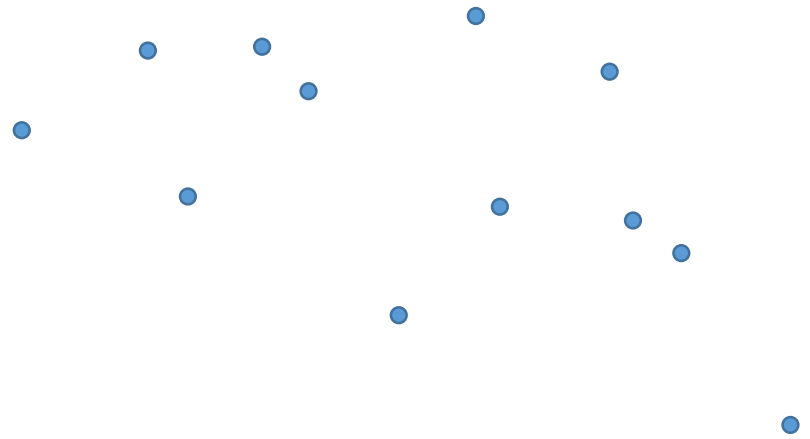
→

A RANDOM set of samples are created

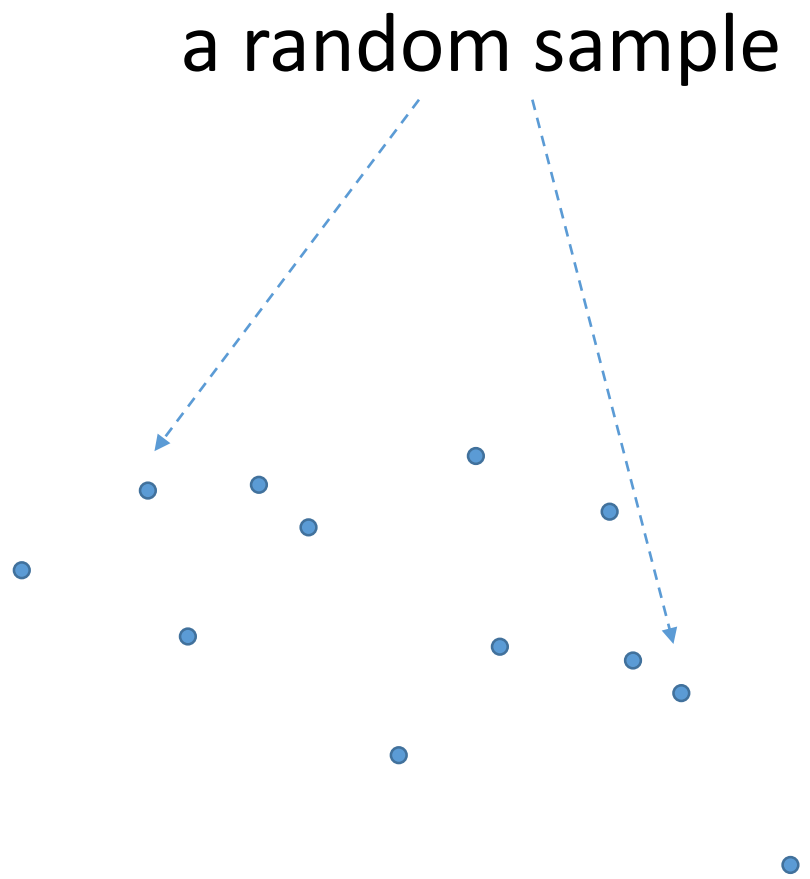
Each random sample proposes the model compatible with all data in the sample

$$S = \{x_1, \ldots, x_s\} \rightarrow m : \begin{cases} f(m, x_1) = 0 \\ \ldots \\ f(m, x_s) = 0 \end{cases}$$

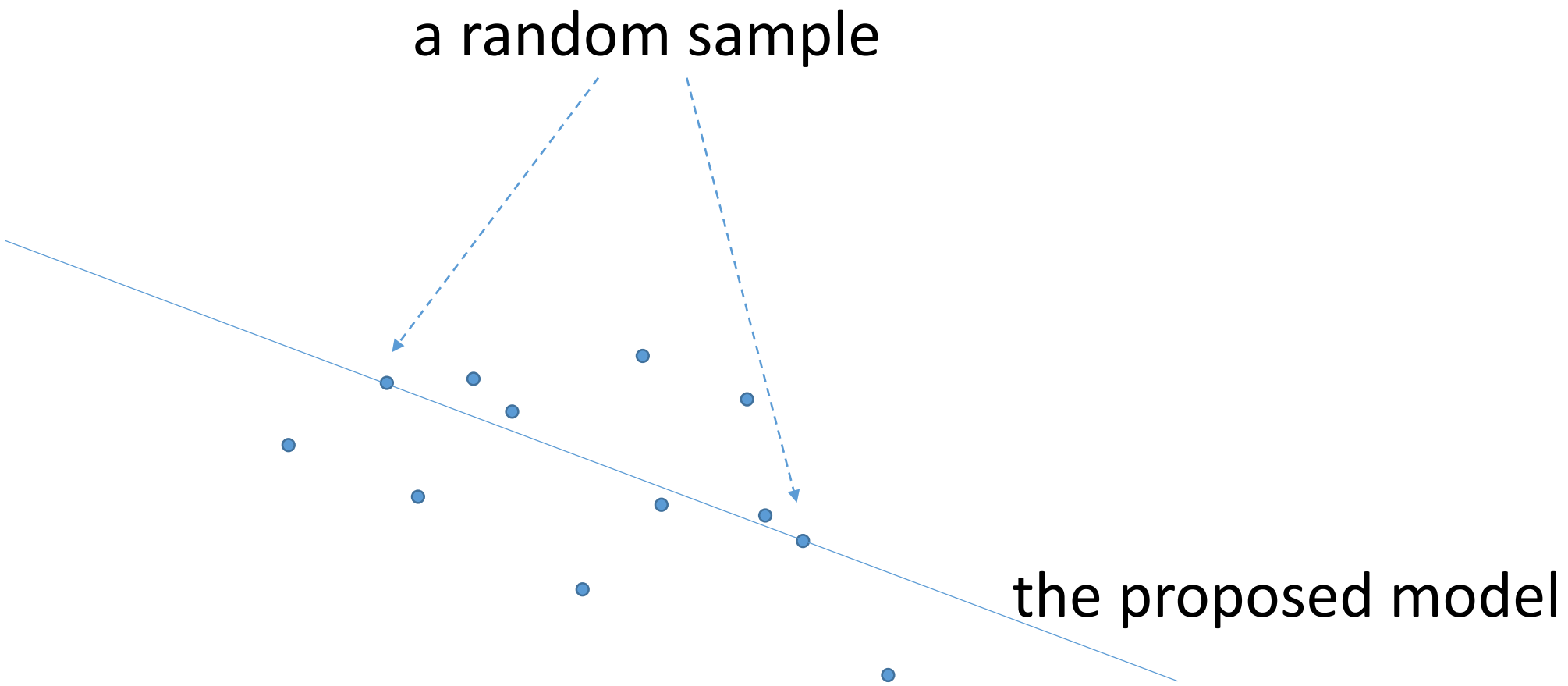
image data



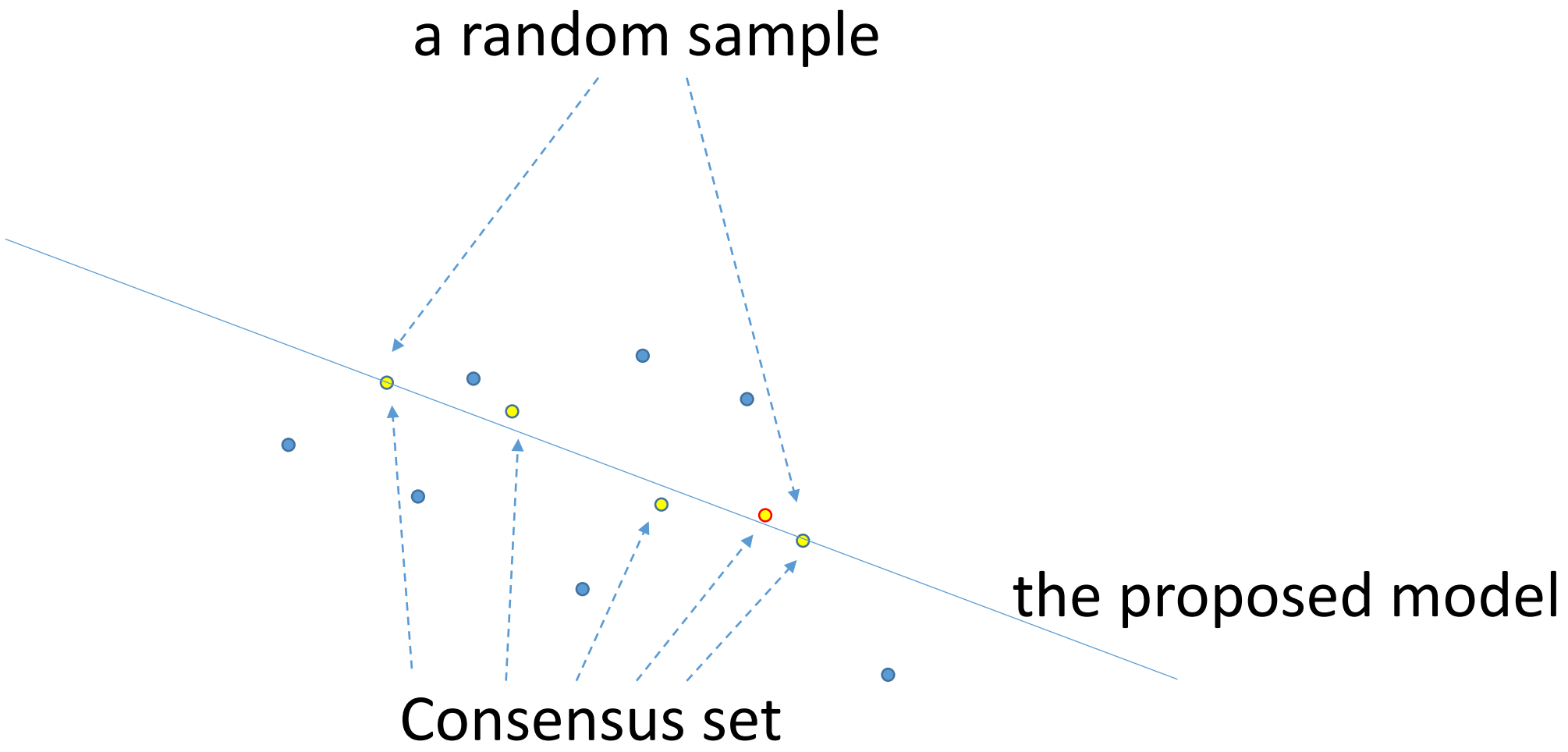
Example with straight lines



A sample and the compatible model



The model proposed by each random sample is voted by other data, that AGREE with it



Consensus: number of data compatible with the proposed model

Consensus set $= \{x_i \in X : |f(m, x_i)| < \epsilon\}$

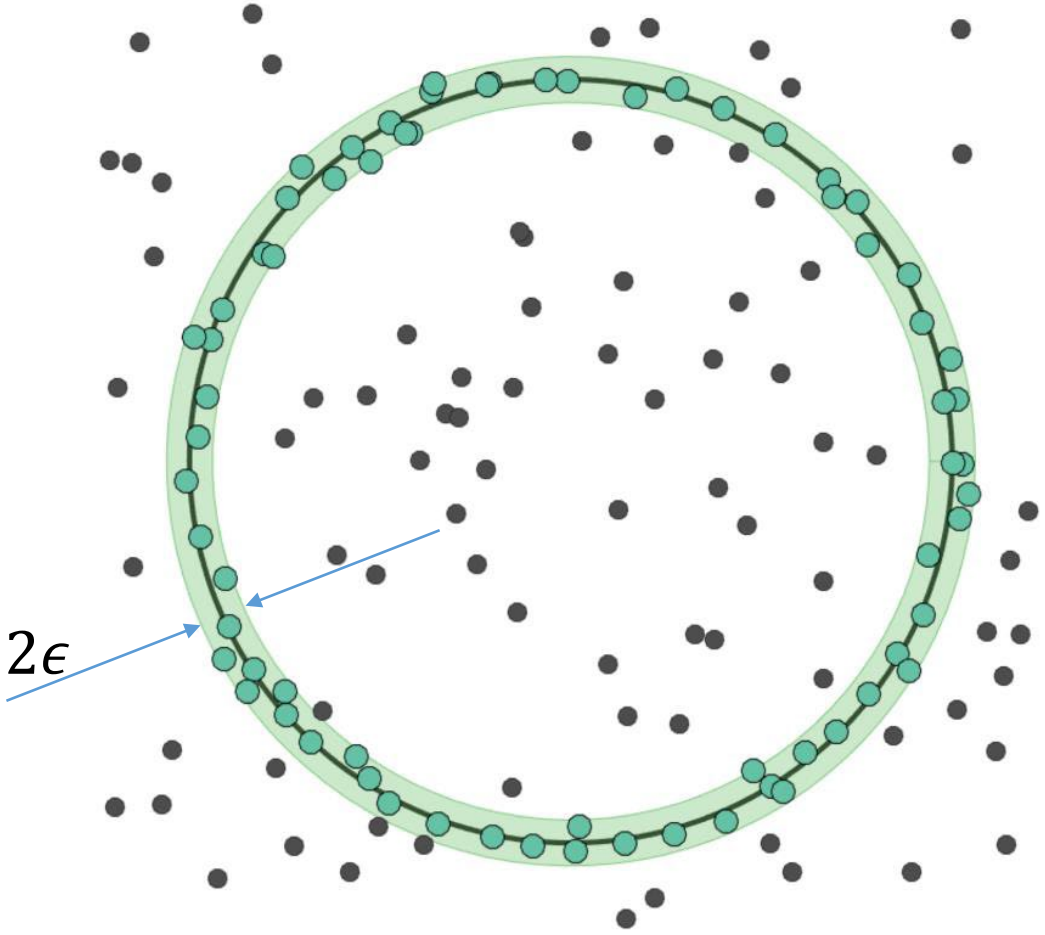
Consensus = $\#\{x_i \in X : |f(m, x_i)| < \epsilon\}$

(notice that is similar to the expression of number of votes to a cell in the Hough Transform)

→ model with maximum Consensus is selected

Consensus set

Example: circumference



Ransac algorithm

Input: X data, ϵ inlier threshold, n maximum iteration numebr

Output: $\hat{m}$ model estimate

```
MAXC = 0;
for(i=1; i<=n; i++){
    randomly select a minimal sample s c X;
    estimate parameters of compatible model m;
    evaluate consensus C = #{x_i ∈ X : |f(m, x_i)| < ε};
    if(C > MAXC){
        MAXC = C;
        m̂ = m;}
Optimize m̂ on its inliers (i.e., its consensus set);
```

Ransac algorithm

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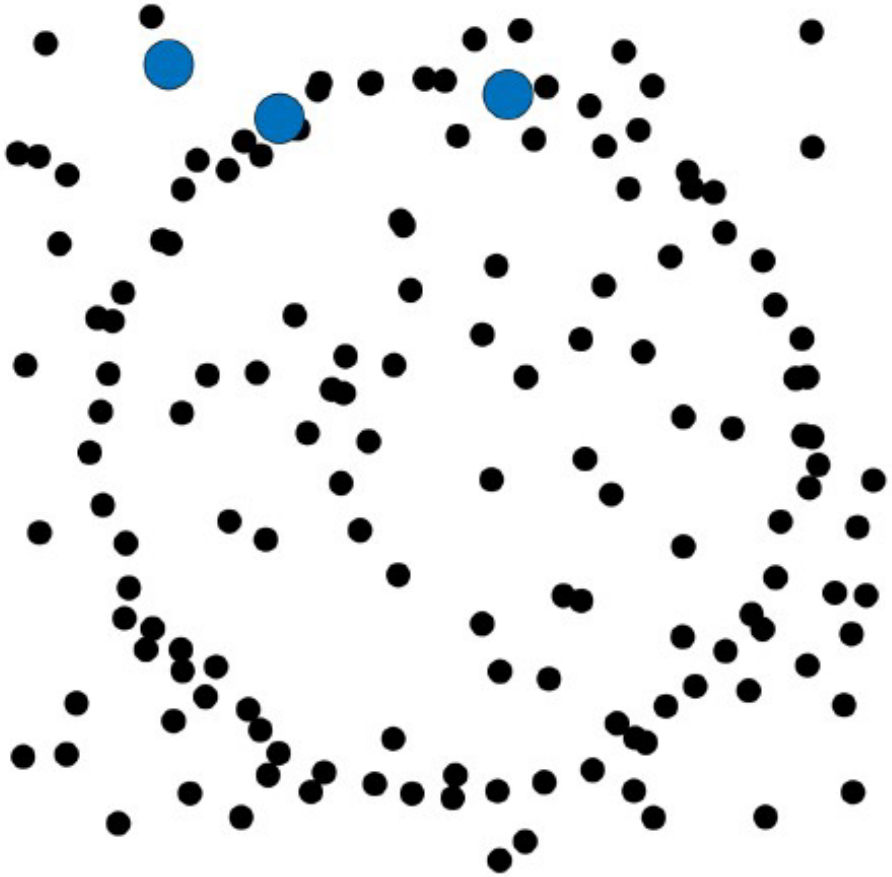
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Optimize $\hat{m}$ on its inliers (i.e., its consensus set);



Ransac algorithm

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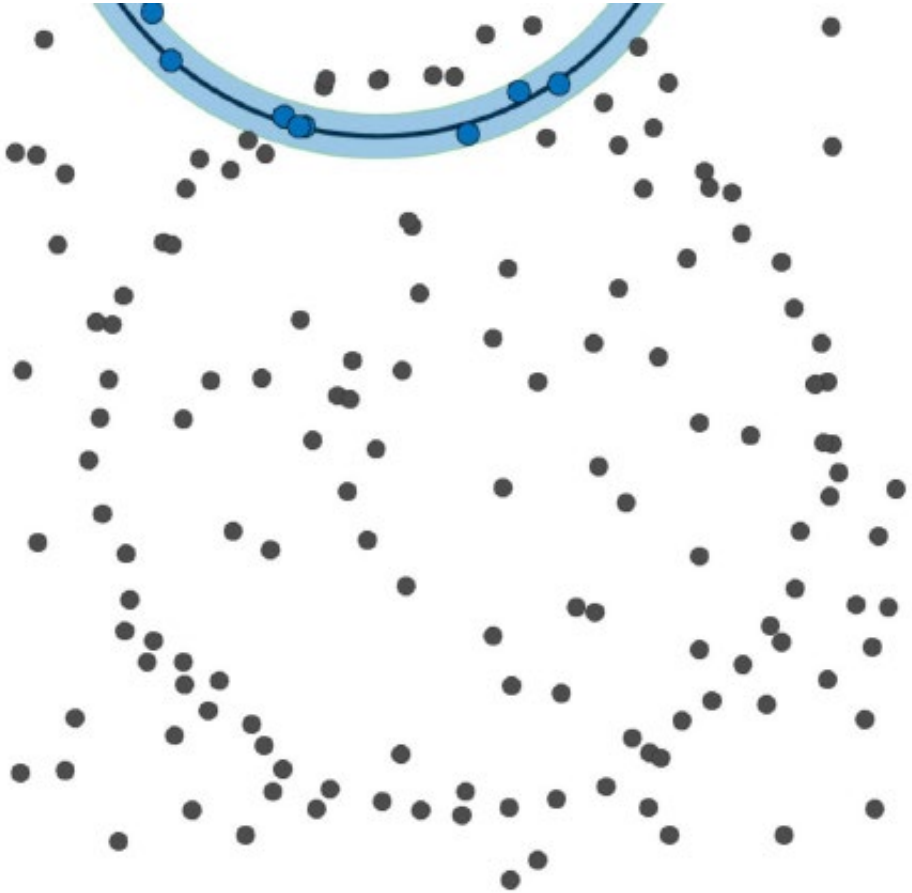
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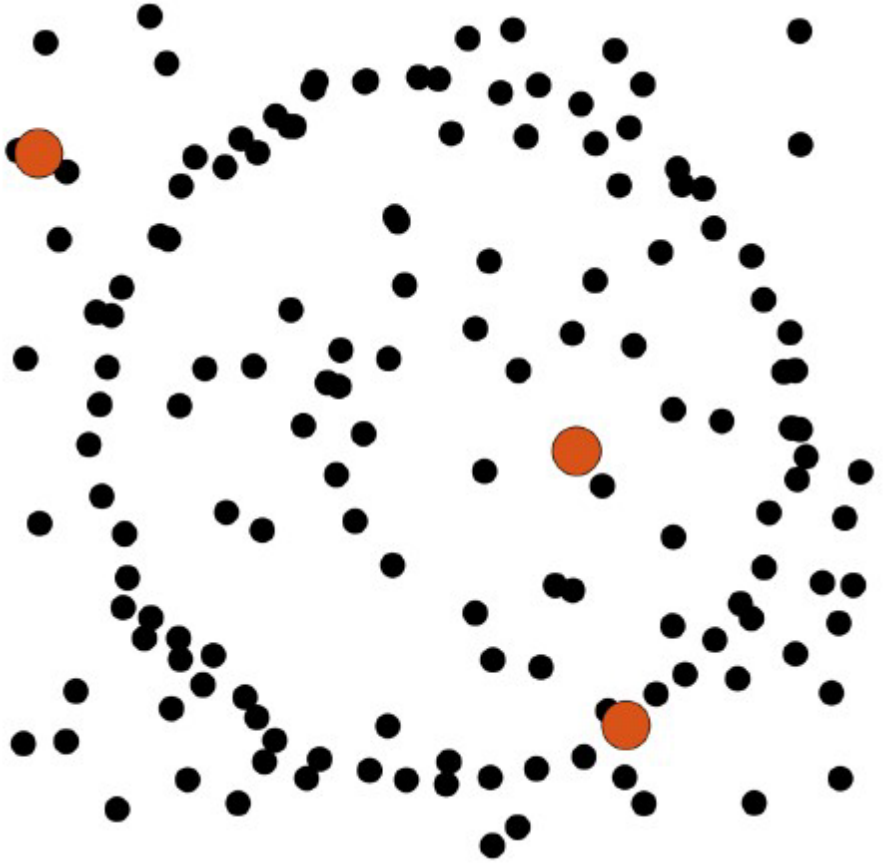
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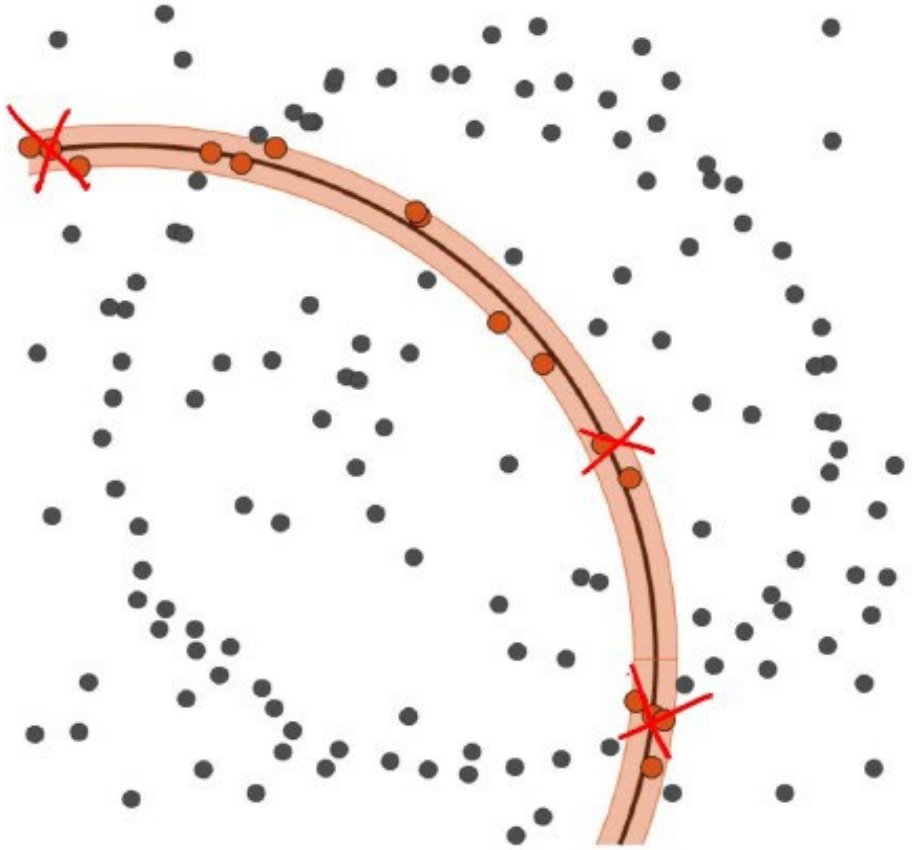
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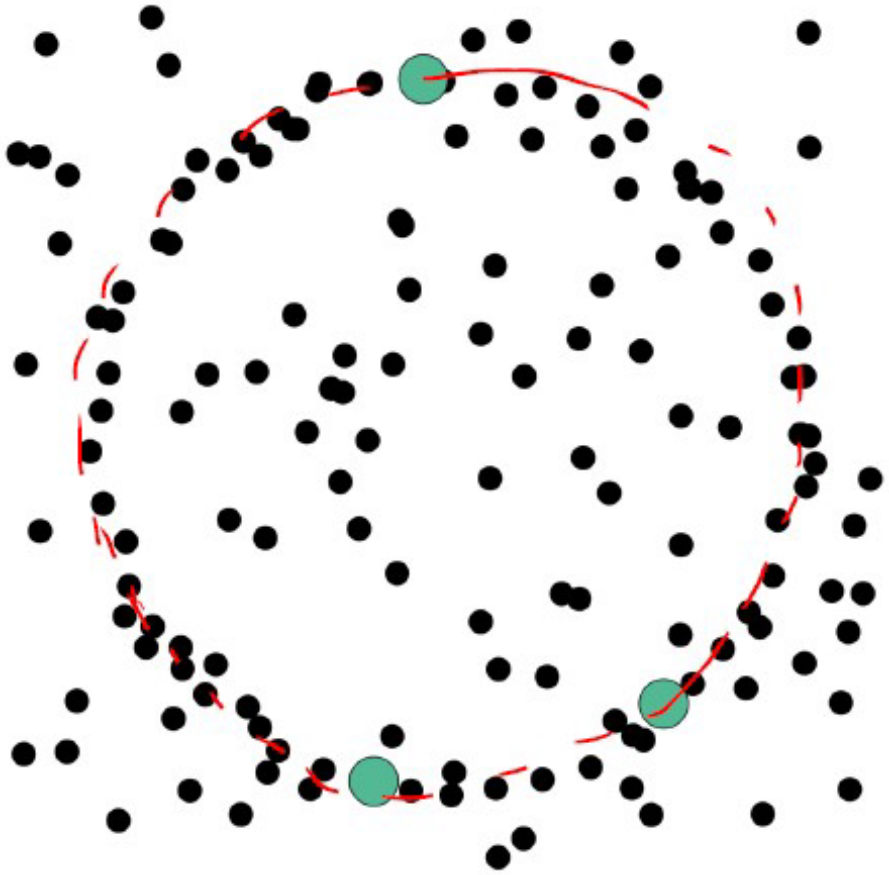
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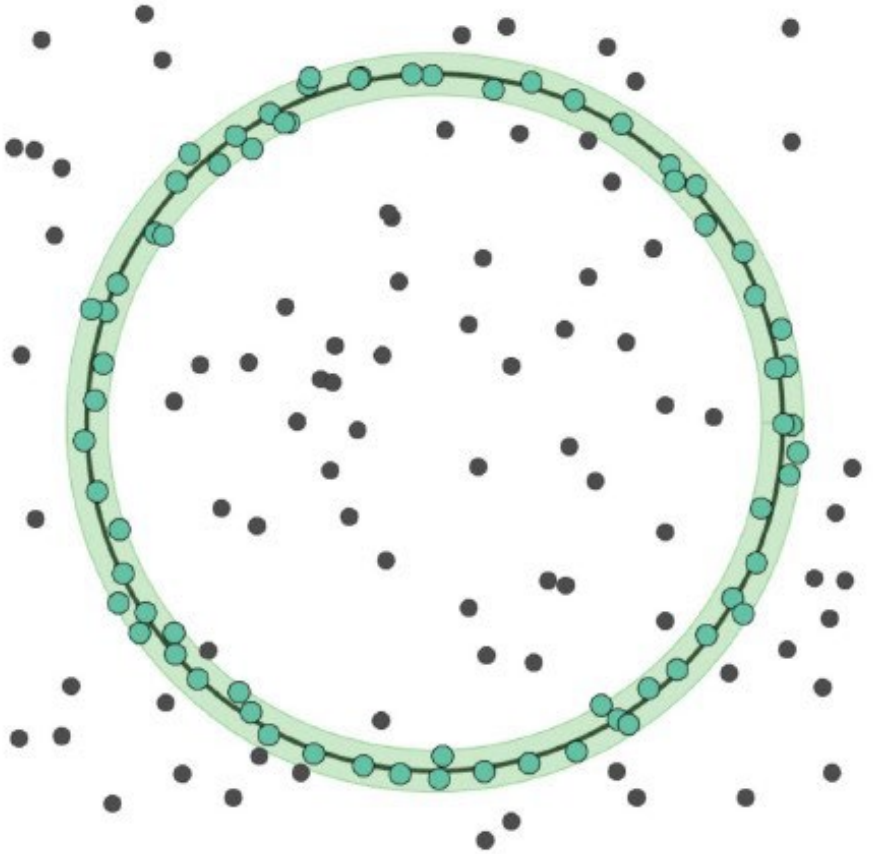
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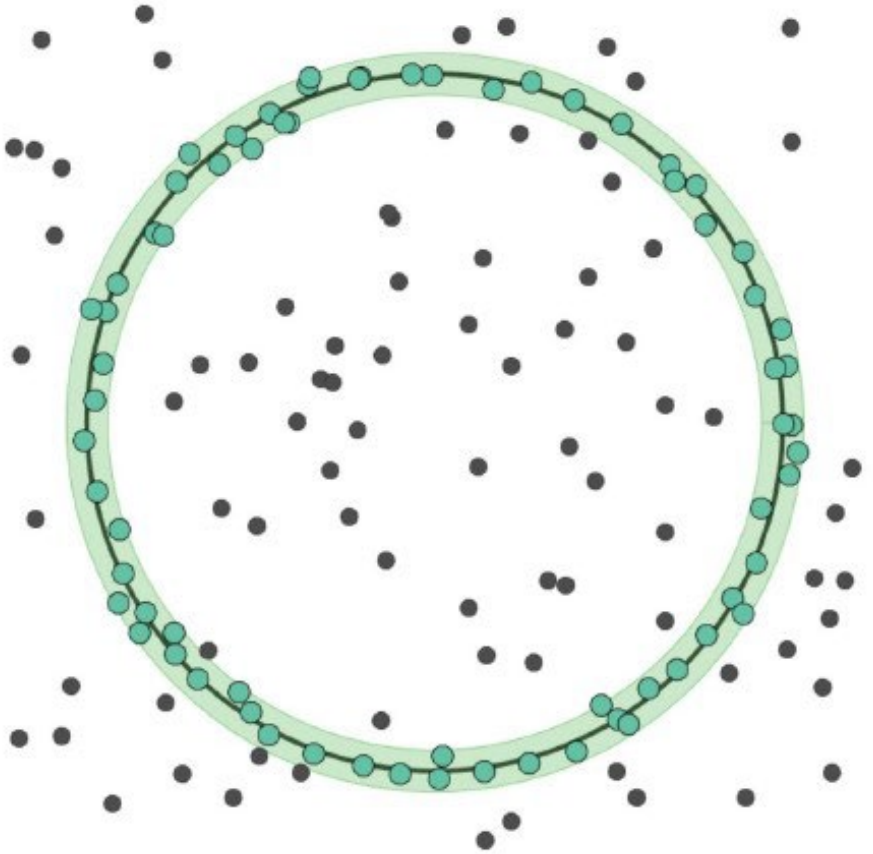
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Optimize $\hat{m}$ on its inliers (i.e., its consensus set);



Observation: random samples are a very small fraction of all the samples: so why does it WORK?

Almost all the samples will contain at least one outlier

BUT

it is sufficient that just ONE sample, among all the generated ones, contains **only** inliers

in order to find the true MODEL present among the data

Probability of failure: ALL the samples must contain at least one outlier

Probability of success: ONE of the samples contains only inliers

How many random samples *n* are needed to have a given probability of success?

- desired probability of success: $1 - \varepsilon$
- (estimated) fraction of inliers: r

$Prob\{good\ sample\} \cong r^s$ (all s data are inliers)

$Prob\{bad\ sample\} = 1 - r^s$ (at least one out of s data is an outlier)

$Prob\{failure\} = (1 - r^s)^n$ (all n samples are bad)

Take n such that $(1 - r^s)^n < \varepsilon$ (in order to have failure prob < ε)

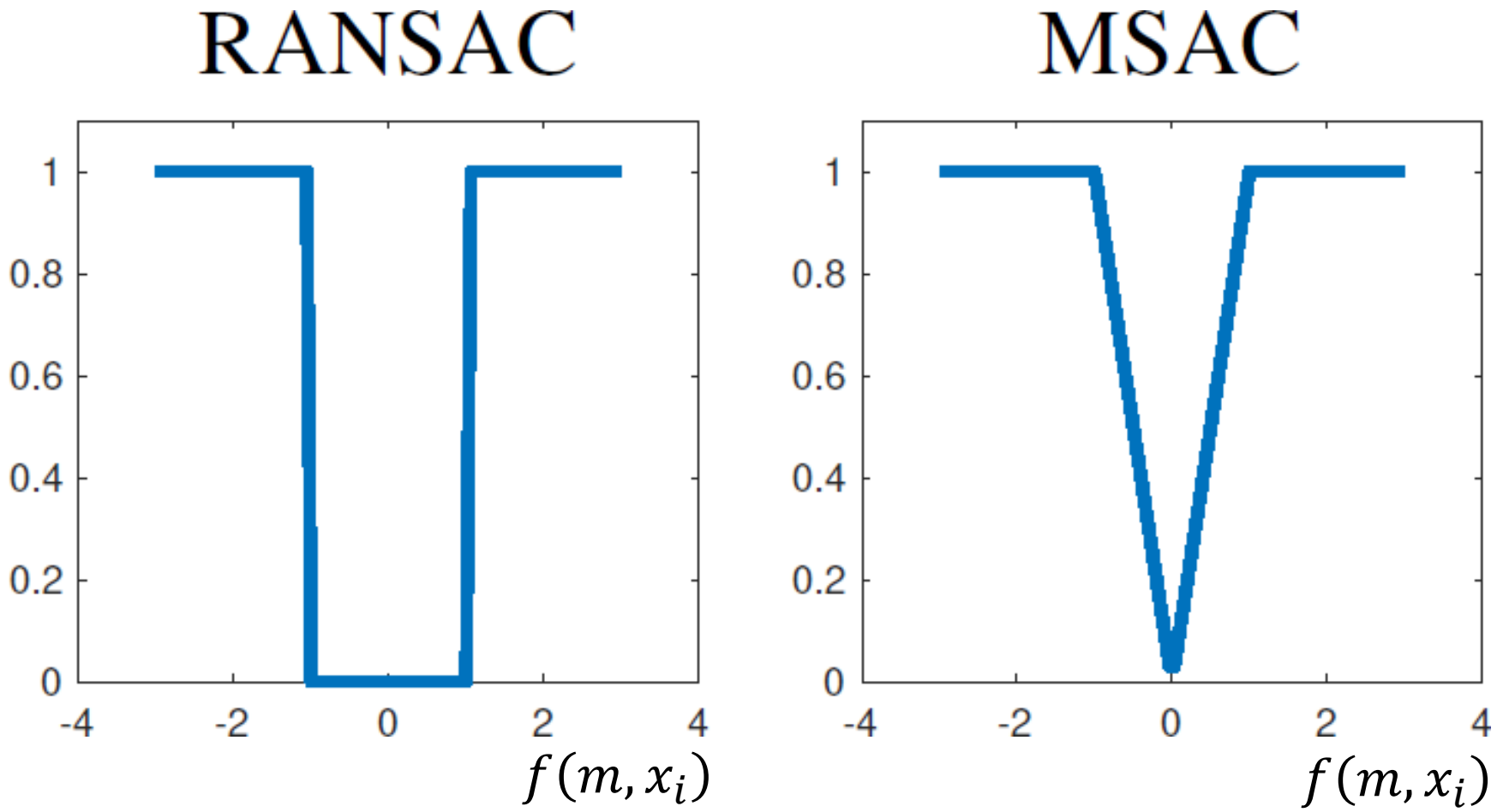
observation

- The probability of success, and thus the number of needed random samples, is independent of the number N of features. It only depends on the percentage of inliers

- Of course the computation time (e.g. in the evaluation of the consensus) depends on the number N of data

Variations: MSAC (Torr & Zisserman 2000)

Minimize the sum of a different function of the residual



Variation: least p-th percentile of squares Rousseeuw e Leroy (1987)

for each random sample,

- estimate the compatible model m
- sort the residuals of the various data $|f(m, x_i)|$ in increasing order
- compute the p-percentile value in the ordered set of residuals*

ouput the model with the minimum p-percentile value

** e.g., p = 50 corresponds to median, p should be < estimated inlier percentage*